COMPLEXITY OF DEEP COMPUTATIONS VIA TOPOLOGY OF FUNCTION SPACES

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Abstract. We use topological methods and a suitable framework to study the complexity of limit computations (such as the "deep computations" arising in recurrent neural networks). By using topology of function spaces —specifically, the classification of Rosenthal compacta— we identify new computational complexity classes. These classes form a strict hierarchy, each level of which is realized by classical computations such as the power method, Bresenham's line algorithm and the Euclidean algorithm, while Newton's method on the Riemann sphere lies outside it; for iterative algorithms, the hierarchy coincides with a hierarchy of tame dynamical systems. The language of model theory, particularly, the concept of *independence* from Shelah's classification theory, provides a medium to translate between topology and computation. Rosenthal compacta, and their measure-theoretic counterpart, give conditions for deep computations to be approximable, deterministically and probabilistically.

Introduction

In this paper we study asymptotic behavior of computations as parameters of the computation tend towards a limit, e.g., the depth of a neural network tending to infinity, or the time interval between layers of a time-series network tending toward zero. Recently, particular cases of this concept have attracted considerable attention in deep learning research (e.g., Neural Ordinary Differential Equations [CRBD], Physics-Informed Neural Networks [RPK19], and Deep Equilibrium Models [BKK]). The formal framework introduced here provides a unified setting to study these limit phenomena from a foundational viewpoint.

In a previous paper [ADIW26], the authors introduced limits of computations, which they called *ultracomputations*, since they arise as ultrafilter limits of standard computations, and *deep computations*, following usage in machine learning [BKK]. There is a technical difference between the two notions, but here we ignore it and speak of deep computations. That paper proved a dichotomy between *stable* (polynomially approximable) and unstable classes of deep computations, using a

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classical criterion of Grothendieck [Gro52]; polynomial approximability of computations corresponds to continuous extendibility in topology, and to *stability* and *type definability* in model theory.

In this paper we refine that dichotomy. An unstable class of deep computations may still be *tame*, in the sense that it does not have Shelah's independence property, or *wild*, in the sense that it does. Within the tame unstable classes we identify three new levels, which give a strict hierarchy of four tame classes below the stable ones, and seven minimal prototypes of tame families. The tools are more recent than those of [ADIW26]: on the topological side, the theory of *Rosenthal compacta* replaces Grothendieck's criterion; on the model-theoretic side, *neostability* replaces stability, and the main notion is the negation of the independence property, commonly called "NIP".

The translation between topology and computation goes through model theory. To every state of a computation we associate its *type*, the list of values of the basic predicates of the model at that state, which describes the logical properties of the state with respect to the rest of the model. The types form a space of real-valued functions, the *space of types*, on which computations act; the idea of embedding models into their type spaces is central in model theory. This embedding lets us use the theory of function spaces, known as C_p*-theory*, to study topological limits of computations. The space of types thus becomes a "Rosetta stone" connecting three classification programs: in topology, the classification of Rosenthal compacta pioneered by Todorčević [Tod99]; in logic, the classification of theories developed by Shelah [She71, She90]; and in statistical learning, VC dimension and PAC learnability [VC71, VC74, Val84]. Under this correspondence, Rosenthal compacta correspond to NIP and to finite VC dimension, which in combinatorics separate polynomial from exponential growth (Section 1.5).

Deep computations arise as topological limits of standard (continuous) computations. The pointwise limits of sequences of continuous functions form the *first Baire class*, one level of topological complexity above the continuous functions; intuitively, functions of Baire class 1 have "controlled" discontinuities. A *Rosenthal compactum* is a compact space of Baire class 1 functions on a Polish (i.e., separable, completely metrizable) space, with the topology of pointwise convergence. Since the late 1970s, Rosenthal compacta have played a central role in functional analysis, especially in the theory of Banach spaces, and Todorčević's trichotomy for them [Tod99] has been used to organize the tame dynamical systems into a hierarchy [GM22]. Todorčević's proof relies on set-theoretic forcing and infinite Ramsey theory; decades later there is still no elementary proof, and his methods continue to be analyzed and exploited [Tod23, HT19]. Beyond the trichotomy, we use the heptachotomy of Argyros, Dodos and Kanellopoulos [ADK08], which identifies seven minimal families of Baire class 1 functions indexed by the binary tree: every relatively compact family indexed by the binary tree contains, along a dyadic subtree, a family equivalent to one of them.

From the point of view of approximation, it is noteworthy that in the settings of Grothendieck [Gro52] and of Bourgain, Fremlin and Talagrand [BFT78], compactness and sequential compactness coincide. Consequently, when the computations are NIP, deep computations, which in principle are abstract ultrafilter limits over a possibly uncountable set, can be approximated by concrete *sequences* of standard computations.

We believe that the results presented in this paper show practitioners of computation, topology, descriptive set theory, or model theory how classification invariants used in their field translate into classification invariants of the other fields. In the interest of accessibility, we do not assume familiarity with advanced topology, model theory, or computing. The only technical prerequisites are undergraduate-level topology and measure theory; the necessary background beyond them is covered in Section 1.

Main results. Theorem 3.1 shows that, when there are countably many predicates, a set of computations satisfies the NIP, feature by feature and on shards, if and only if its deep computations are of the first Baire class on each shard, that is, form Rosenthal compacta; in that case every deep computation is the limit of a *sequence* of computations. Using Todorčević's trichotomy, we then define three levels of tameness (Definition 3.2) and obtain the hierarchy

$$\text{stable} \Rightarrow \text{NIP}_3 \Rightarrow \text{NIP}_2 \Rightarrow \text{NIP}_1 \Rightarrow \text{NIP},$$

in which every implication is strict. Section 5 realizes each level by a classical algorithm: a rotation is stable; the power method is NIP_3 but not stable; Bresenham's line algorithm in the plane is NIP_2 but not NIP_3, and in space NIP_1 but not NIP_2; the Euclidean algorithm, acting on the projective line by the group that its steps generate, is NIP but not NIP_1, and its deep computations include exact equality tests; and Newton's method on the Riemann sphere is not NIP. At the level NIP but not NIP_1 the computations form a group with two generators; whether a single iterated map can reach this level is an open question, closely related to a question of Glasner and Megrelishvili (Question 7.2). For iterative algorithms the hierarchy coincides with the hierarchy of tame dynamical systems of Glasner and Megrelishvili [GM22]. The heptachotomy of Argyros, Dodos and Kanellopoulos then shows that every tame family of computations indexed by the binary tree contains, along a dyadic subtree, one of seven minimal families, which we read as primitives of binary search (Section 3.3). Finally, Theorem 6.7 and Corollary 6.14 relate NIP and Talagrand stability to Monte Carlo computability of deep computations.

Organization of the paper. In Section 1, we present the topological and combinatorial preliminaries, including five canonical Rosenthal compacta (Section 1.4) and the role of NIP in model theory, combinatorics and learning (Section 1.5). In Section 2, we introduce compositional computation structures, the structural and model-theoretic viewpoint (no previous exposure to model theory is needed), together with stability; the section ends with two motivating examples, Newton's method for square roots and Newton's method on the Riemann sphere, whose different behaviors the classification is designed to separate (Section 2.5). Section 3 is devoted to the classification of deep computations; it contains the main theorems. Section 4 works out Newton's method in detail, and Section 5 places classical algorithms at every level of the hierarchy and beyond it. Section 6 replaces topological limits by a probabilistic viewpoint, which we intend to develop further in future research. Section 7 collects open questions.

Throughout the paper, our results pertain to classical models of computation (particularly computations involving real-valued quantities that are known and manipulated to a finite degree of precision).

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1. Topological preliminaries: From continuity to Baire class 1

In this section we recall notions and results from general topology and function space theory. For simplicity, we assume that all spaces are Hausdorff unless explicitly stated otherwise.

A subset of a topological space is F_σ if it is a countable union of closed sets, and G_δ if it is a countable intersection of open sets. In a metric space, every open set is F_σ; equivalently, every closed set is G_δ.

A *Polish space* is a separable and completely metrizable topological space, i.e., one admitting a complete metric that induces its topology. The class of Polish spaces is closed under countable topological products. Classical examples of Polish spaces (besides the real line) include the Cantor space $2^{\mathbb{N}}$ (the set of all infinite binary sequences, endowed with the product topology), the Baire space $\mathbb{N}^{\mathbb{N}}$ (the set of all infinite sequences of naturals, also with the product topology), and the space $\mathbb{R}^{\mathbb{N}}$ of sequences of real numbers.

Fact 1.1. [Eng89, 4.3.24] *A subset of a Polish space is itself Polish in the subspace topology if and only if it is a G_δ set. In particular, closed subsets and open subsets of Polish spaces are also Polish spaces.*

If X is a topological space and A is a subset of X, then A is said to be *nowhere dense* if its closure has an empty interior, and of the *first category* if A is a countable union of closed nowhere dense sets; otherwise, A is said to be of the *second category*. Sets $A \subseteq X$ of the first category are also called *meager*, and their complements $X \setminus A$ are called *comeager*. Thus, the set of meager subsets of X is the σ-ideal generated by the nowhere dense subsets of X. A topological space X is *Baire* if every comeager subset of X is dense. Equivalently, X is Baire if any countable intersection of open dense sets remains dense. The Baire Category Theorem states that compact Hausdorff spaces and all completely metrizable spaces (hence, all Polish spaces) are Baire.[1]

Given two topological spaces X and Y we denote by $\mathrm{C_p}(X, Y)$ the set of all continuous functions $f : X \to Y$ endowed with the topology of pointwise convergence, i.e., with the relative (subspace) product topology $\mathrm{C_p}(X, Y) \subseteq Y^X$—where Y carries its topology, but X serves merely as a (discrete) index set stripped of any topology. The space $\mathrm{C_p}(X, \mathbb{R})$ of continuous real-valued functions on X is denoted simply $\mathrm{C_p}(X)$.

A function $f \in Y^X$ between topological spaces X, Y is said to be of *Baire class 1*, or a *Baire-1* function, if there is a sequence $(f_n)_{n\in\mathbb{N}} \subseteq \mathrm{C_p}(X, Y)$ such that $f = \lim_{n\to\infty} f_n$ (i.e., $f(x) = \lim_{n\to\infty} f_n(x)$ for all $x \in X$). An elementary argument shows that Baire-1 functions are continuous everywhere except on a set of the first category. If X and Y are topological spaces, the space of Baire-1 functions $f : X \to Y$ endowed with the topology of pointwise convergence is denoted $\mathrm{B_1}(X, Y)$; thus, $\mathrm{C_p}(X, Y) \subseteq \mathrm{B_1}(X, Y) \subseteq Y^X$ are topological inclusions. We abbreviate $\mathrm{B_1}(X, \mathbb{R})$ as $\mathrm{B_1}(X)$. Relatively compact subsets of $\mathrm{C_p}(X)$ and of $\mathrm{B_1}(X)$ (particularly for X Polish) are of interest in analysis and topological dynamics.

A topological space X is *perfectly normal* if it is normal and every closed subset of X is a G_δ set (equivalently, every open subset of X is an F_σ set). Every metrizable space (hence, every Polish space) is perfectly normal.

[1] More generally, any Čech-complete space is Baire [Eng89, Theorem 3.9.3].

The following fact was proved by Baire in his groundbreaking thesis. For a proof, see [Tod97, Section 10].

Fact 1.2 (Baire)**.** *If X is perfectly normal, then the following conditions are equivalent for a function $f : X \to \mathbb{R}$:*

(1) *f is a Baire class 1 function, that is, f is a pointwise limit of continuous functions.*
(2) *$f^{-1}[U]$ is an F_σ subset of X whenever $U \subseteq \mathbb{R}$ is open.*

If X is Baire, then (1) and (2) are equivalent to:

(3) *For every closed $F \subseteq X$, the restriction $f|_F$ has a point of continuity.*

Moreover, if X is Polish and $f \notin \mathrm{B}_1(X)$, then there exist countable $D_0, D_1 \subseteq X$ and reals $a < b$ such that

$$D_0 \subseteq f^{-1}(-\infty, a], \quad D_1 \subseteq f^{-1}[b, \infty), \quad \overline{D_0} = \overline{D_1}.$$

1.1. **From Rosenthal's dichotomy to the Bourgain–Fremlin–Talagrand dichotomy to Shelah's NIP.** In metrizable spaces, the closure of a subset consists of limits of convergent sequences therein. In general, such a characterization of the closure fails in remarkable ways.

The n-th coordinate map $p_n : 2^{\mathbb{N}} \to \{0, 1\}$ on the Cantor space $X = 2^{\mathbb{N}}$ $(= \{0, 1\}^{\mathbb{N}})$ is continuous for each $n \in \mathbb{N}$; however, $(p_n)_{n\in\mathbb{N}}$ has *no* convergent subsequences in $\mathbb{R}^X$ ([Tod97, Chapter 1.1]). In a sense, this example exhibits the worst failure of sequential convergence possible: The closure of $\{p_n\}$ in $\{0, 1\}^X$ (or in $\mathbb{R}^X$ for that matter) is homeomorphic to the Stone–Čech compactification $\beta\mathbb{N}$ of the (discrete) space of natural numbers. The space $\beta\mathbb{N}$ has cardinality 2^{2^ω} (largest among all separable topological spaces) and, effectively, is the set of ultrafilters on $\mathbb{N}$.[2]

The following theorem, proved by Haskell Rosenthal in 1974, is fundamental in functional analysis and captures a sharp division in the behavior of sequences in a Banach space.

Theorem 1.3 (Rosenthal's Dichotomy, [Ros74])**.** *If X is Polish and $(f_n)_{n\in\mathbb{N}} \subseteq \mathrm{C}_\mathrm{p}(X)$ is pointwise bounded, then (f_n) has a convergent subsequence, or a subsequence whose closure in $\mathbb{R}^X$ is homeomorphic to $\beta\mathbb{N}$.*

The genesis of the dichotomy was *Rosenthal's ℓ_1-Theorem*, which states that a Banach space includes an isomorphic copy of ℓ_1 (the space of absolutely summable sequences), or else every bounded sequence therein is weakly Cauchy. The ℓ_1-Theorem connects diverse areas: Banach space geometry, Ramsey theory, set theory, and topology of function spaces.

As we move from $\mathrm{C}_\mathrm{p}(X)$ to the larger space $\mathrm{B}_1(X)$, a dichotomy paralleling the ℓ_1-Theorem holds. This result is usually not phrased as a dichotomy in the above manner, but rather as an equivalence (see Theorem 1.4 below).

First, we introduce some useful notation. For any $f \in \mathbb{R}^X$ and any real a, define

$$X^f_{\leq a} \coloneqq f^{-1}(-\infty, a], \qquad X^f_{\geq a} \coloneqq f^{-1}[a, +\infty).$$

[2]Our earlier proof [ADIW26] of the existence of deep equilibria in compositional computation structures relies on a classical result on the existence of idempotent elements of $\beta\mathbb{N}$.

For any set $A \subseteq \mathbb{R}^X$ and any real a, define (by a slight abuse of notation)

$$X^A_{\leq a} \coloneqq \bigcap_{f\in A} X^f_{\leq a} = \{x \in X : f(x) \leq a \text{ for all } f \in A\},$$
$$X^A_{\geq a} \coloneqq \bigcap_{f\in A} X^f_{\geq a} = \{x \in X : f(x) \geq a \text{ for all } f \in A\}.$$

(In case $A = \emptyset$, we define $X^\emptyset_{\geq a} = X = X^\emptyset_{\leq a}$.) For any sequence $\{f_n\} \subseteq \mathbb{R}^X$ and $I \subseteq \mathbb{N}$, define $I^{\complement} \coloneqq \mathbb{N} \setminus I$ and $f_I \coloneqq \{f_i : i \in I\}$.

Theorem 1.4 ("The BFT Dichotomy". Bourgain–Fremlin–Talagrand [BFT78])**.** *Let X be a Polish space and $A \subseteq \mathrm{C_p}(X)$ be pointwise bounded. The following are equivalent:*

(i) A is relatively compact in $\mathrm{B}_1(X)$.
(ii) For every $(f_n)_{n\in\mathbb{N}} \subseteq A$ and every $a < b$ there is $I \subseteq \mathbb{N}$ such that

$$X^{f_I}_{\leq a} \cap X^{f_{I^{\complement}}}_{\geq b} = \emptyset.$$

(As stated above, the BFT Dichotomy is a particular case of the equivalence (ii)⇔(v) in [BFT78, Corollary 4G].)

The sets $X^{f_I}_{\leq a}$ and $X^{f_{I^{\complement}}}_{\geq b}$ appearing in condition Theorem 1.4*(ii)* are defined, respectively, in terms of $|I|$-many inequalities of the form $f_i(x) \leq a$, and $|I^{\complement}|$-many of the form $f_j(x) \geq b$. Thus, at least one of $X^{f_I}_{\leq a}$ and $X^{f_{I^{\complement}}}_{\geq b}$ is defined by the satisfaction of infinitely (countably) many inequalities. For our purposes, it is more natural to consider only finitely many inequalities at a time, which motivates the definitions below.

Definition 1.5. We say that a collection of functions $A \subseteq \mathbb{R}^X$ *has the finitary No-Independence Property (NIP)* if, for all sequences $(f_n)_{n\in\mathbb{N}} \subseteq A$ and reals $a < b$, there exist finite disjoint sets $E, F \subseteq \mathbb{N}$ such that $X^{f_E}_{\leq a} \cap X^{f_F}_{\geq b} = \emptyset$. We say that such E, F *witness finitary NIP for* (f_n) *and* a, b.

A set $A \subseteq \mathbb{R}^X$ *has the finitary Independence Property (IP)* if it does not have finitary NIP, i.e., if there exists a sequence $(f_n)_{n\in\mathbb{N}} \subseteq A$ and reals $a < b$ such that for every pair of finite disjoint sets $E, F \subseteq \mathbb{N}$, we have $X^{f_E}_{\leq a} \cap X^{f_F}_{\geq b} \neq \emptyset$.

If the word "finite" is omitted in the above definitions, we obtain the definitions of *countable* NIP (weaker than finitary NIP) and *countable* IP (stronger than finitary IP), respectively.

If we insist on witnesses $E, F \subseteq \mathbb{N}$ such that $F = E^{\complement}$, we call the respective properties "BFT-NIP" and "BFT-IP". Thus, Theorem 1.4 becomes (for pointwise bounded function collections $A \subseteq \mathrm{C_p}(X)$) that A is relatively compact in $\mathrm{B}_1(X)$ if and only if A has BFT-NIP.

Unless otherwise specified, IP/NIP shall mean *finitary* IP/NIP henceforth.

Theorem 1.6. *Let X be a compact metrizable (hence Polish) space. For every pointwise bounded $A \subseteq \mathrm{C_p}(X)$, the following properties are all equivalent:*

(i) A is relatively compact in $\mathrm{B}_1(X)$;
(ii) A has BFT-NIP;
(iii) A has countable NIP;
(iv) A has finitary NIP.

(The equivalence *(ii)⇔(iv)* holds for arbitrary compact X, without assuming pointwise boundedness. The equivalence *(ii)⇔(iii)*, for arbitrary X.)

Proof. (i)⇔(ii) is Theorem 1.4. The implications (iv)⇒(iii) and (ii)⇒(iii) are trivial, and (iii)⇒(ii) follows by taking $I = E$, since then $F \subseteq I^{\complement}$. For (ii)⇒(iv), suppose that (f_n) and $a < b$ witness finitary IP and fix $I \subseteq \mathbb{N}$. The closed sets $X^{f_n}_{\leq a}$ $(n \in I)$ and $X^{f_m}_{\geq b}$ $(m \in I^{\complement})$ have the finite intersection property, so by compactness $X^{f_I}_{\leq a} \cap X^{f_{I^{\complement}}}_{\geq b} \neq \emptyset$. □

Theorem 1.6 may be stated as the following dichotomy (under the assumptions) for any compact metrizable X: either A is relatively compact in $\mathrm{B}_1(X)$, or A has IP (in either sense).

The Independence Property was first isolated by Saharon Shelah in model theory as a dividing line between theories whose models are "tame" (corresponding to NIP) and theories whose models are "wild" (corresponding to IP). See [She71, Definition 4.1], [She90]. We discuss this dividing line in Section 1.5.

1.2. **Rosenthal compacta.** The dichotomy of Theorem 1.4 led Godefroy [God80] to the following definition.

Definition 1.7. A Rosenthal compactum is any topological space realized as a compact subset of the space $\mathrm{B}_1(X) = \mathrm{B}_1(X, \mathbb{R})$ (equipped with the topology of pointwise convergence) of all real functions of the first Baire class on some Polish space X.

A Rosenthal compactum K is necessarily Hausdorff since it is a topological subspace of the Hausdorff product space $\mathbb{R}^X$.

Rosenthal compacta possess significant topological and dynamical tameness properties, and play an important role in functional analysis, measure theory, dynamical systems, descriptive set theory, and model theory. In this paper, we use them to study deep computations.

1.3. **The special case $\mathrm{B}_1(X, \mathbb{R}^{\mathcal{P}})$ with $\mathcal{P}$ countable.** Fix an arbitrary (at most) countable set $\mathcal{P}$ whose elements $P \in \mathcal{P}$ will be called *predicate symbols* or *formal predicates*. Our present goal is to characterize relatively compact subsets of $\mathrm{B}_1(X, \mathbb{R}^{\mathcal{P}})$, where X is always assumed to be a perfectly normal space (typically a Polish space).

The set $\mathcal{P}$ shall be considered discrete whenever regarded as a topological space. Since $\mathrm{C}_{\mathrm{p}}(X, \mathbb{R}^{\mathcal{P}}) \subseteq \mathrm{B}_1(X, \mathbb{R}^{\mathcal{P}}) \subseteq (\mathbb{R}^{\mathcal{P}})^X$, the "ambient" space $(\mathbb{R}^{\mathcal{P}})^X$ is quite relevant. The product $X \times \mathcal{P}$ will be regarded as either a point set, or as a topological product depending on context. We have natural homeomorphic identifications

$$(\mathbb{R}^{\mathcal{P}})^X \cong \mathbb{R}^{X \times \mathcal{P}} \cong (\mathbb{R}^X)^{\mathcal{P}},$$

given by

$$\mathbb{R}^{X \times \mathcal{P}} \to (\mathbb{R}^{\mathcal{P}})^X : \varphi \mapsto \hat{\varphi}$$
$$\mathbb{R}^{X \times \mathcal{P}} \to (\mathbb{R}^X)^{\mathcal{P}} : \varphi \mapsto \varphi\hat{},$$

where

$$\hat{\varphi}(x) \coloneqq \varphi(x, \cdot) \in \mathbb{R}^{\mathcal{P}}, \qquad \varphi\hat{}(P) \coloneqq \varphi(\cdot, P) \in \mathbb{R}^X.$$

Such identifications view X and $\mathcal{P}$ as mere point sets (the topology of X in particular plays no role).

For $x \in X$, define the "left projection" map

$$\lambda_x : \mathbb{R}^{X\times\mathcal{P}} \to \mathbb{R}^{\mathcal{P}} : \varphi \mapsto \lambda_x(\varphi) \coloneqq \hat{\varphi}(x);$$

for $P \in \mathcal{P}$, the "right projection" map

$$\rho_P : \mathbb{R}^{X\times\mathcal{P}} \to \mathbb{R}^{X} : \varphi \mapsto \rho_P(\varphi) \coloneqq \varphi\hat{(}P).$$

For fixed $x \in X$ and $P \in \mathcal{P}$, we also have canonical projection maps

$$\lambda_x : \mathbb{R}^X \to \mathbb{R} : f \mapsto f(x), \qquad \rho_P : \mathbb{R}^{\mathcal{P}} \to \mathbb{R} : f \mapsto f(P).$$

When clear from context, rather than using the specific symbols ("λ" for left,"ρ" for right) to denote projections, we may use the generic symbol "π"; thus, π_x may mean λ_x, and π_P may mean ρ_P.

The Proposition below reduces the study of $\mathbb{R}^{\mathcal{P}}$-valued continuous or Baire-1 functions on X to the special case of real-valued ones.

Proposition 1.8. *The identification* $(\mathbb{R}^{\mathcal{P}})^X \cong \mathbb{R}^{X\times\mathcal{P}} \cong (\mathbb{R}^X)^{\mathcal{P}}$ *induces identifications*

$$\mathrm{C_p}(X, \mathbb{R}^{\mathcal{P}}) \cong \mathrm{C_p}(X\times\mathcal{P}) \cong \mathrm{C_p}(X)^{\mathcal{P}}, \quad \mathrm{B}_1(X, \mathbb{R}^{\mathcal{P}}) \cong \mathrm{B}_1(X\times\mathcal{P}) \cong \mathrm{B}_1(X)^{\mathcal{P}}.$$

(The cardinality of $\mathcal{P}$ plays no role.)

Proof. The identification of $\mathrm{C_p}$-spaces follows trivially from the definition of topological product and the fact that $\mathcal{P}$ is discrete: a continuous map $X \to \mathbb{R}^{\mathcal{P}}$ is precisely a $\mathcal{P}$-indexed family of continuous functions $X \to \mathbb{R}$, and these correspond to continuous functions $X \times \mathcal{P} \to \mathbb{R}$. The identification of Baire-1 spaces follows immediately, since it is defined in terms of the purely topological notion of limit (in the ambient space) of sequences of continuous functions. □

Given $A \subseteq Y^X$ and $K \subseteq X$ we write $A|_K \coloneqq \{f|_K : f \in A\}$, i.e., the set of all restrictions of functions in A to K. The following Theorem is a slightly more general version of Theorem 1.4.

Theorem 1.9. *Assume that* $\mathcal{P}$ *is countable,* X *is a Polish space, and* $A \subseteq \mathrm{C_p}(X, \mathbb{R}^{\mathcal{P}})$ *is pointwise bounded in the sense that* $\pi_P \circ A$ *(*$\subseteq \mathrm{C_p}(X)$*) is pointwise bounded for every* $P \in \mathcal{P}$*. The following are equivalent for every compact* $K \subseteq X$*:*

(i) $A|_K$ *is relatively compact in* $\mathrm{B}_1(K, \mathbb{R}^{\mathcal{P}})$*;*
(ii) $\pi_P \circ A|_K$ *has NIP for every* $P \in \mathcal{P}$*.*

Proof. By Proposition 1.8 and Tychonov's theorem, (i) holds if and only if $\pi_P \circ A|_K$ is relatively compact in $\mathrm{B}_1(K)$ for every $P \in \mathcal{P}$. Since K is compact metrizable, Theorem 1.6 shows that this is equivalent to (ii). □

We conclude this section with an elementary but important observation.

Lemma 1.10. *Assume that* X *is Hausdorff and that* $A \subseteq \mathrm{C_p}(X)$*. The following are equivalent for every* $L \subseteq X$*:*

(i) $A|_L$ *satisfies the NIP;*
(ii) $A|_{\overline{L}}$ *satisfies the NIP.*

Proof. (ii)⇒(i) is trivial. Conversely, if $(f_n) \subseteq A$ and $a < b$ witness the IP on $\overline{L}$, fix $a < a' < b' < b$. For finite disjoint $E, F \subseteq \mathbb{N}$, the open set $\bigcap_{n\in E} f_n^{-1}(-\infty, a') \cap \bigcap_{n\in F} f_n^{-1}(b', \infty)$ meets $\overline{L}$, hence meets L; so (f_n) and $a' < b'$ witness the IP on L. □

1.4. **Five canonical Rosenthal compacta.** Five Rosenthal compacta, all built on the Cantor set, are the canonical members of the classes in Todorčević's trichotomy (Theorem 3.3). By the trichotomy, the space of deep computations of each non-metrizable NIP example of Section 5 contains a copy of one of them. We fix their notation here. Recall that a topological space is *separable* if it has a countable dense subset, *hereditarily separable* if every subspace is separable, and *first countable* if every point has a countable base of neighborhoods.

Let $2^{\mathbb{N}}$ carry the product topology and the lexicographic order $<$, whose order topology is the product topology. For a finite word $t \in 2^{<\mathbb{N}}$ we write $|t|$ for its length, and for $a \in 2^{\mathbb{N}}$ we write $a{\restriction}n$ for the word formed by the first n bits of a. Two words are *comparable* if one extends the other, and an *antichain* is a set of pairwise incomparable words. For a finite word $t \in 2^{<\mathbb{N}}$, let $t^\frown 0^\infty$ and $t^\frown 1^\infty$ be the sequences that begin with t and continue with all 0's, respectively all 1's; the set of sequences extending t is the clopen interval $[t^\frown 0^\infty, t^\frown 1^\infty]$. We use the following functions $2^{\mathbb{N}} \to \{0, 1\}$:

- the *prefix test* v_t, the characteristic function of the set of sequences extending t;
- the *equality test* δ_a, the characteristic function of $\{a\}$, for $a \in 2^{\mathbb{N}}$;
- the *threshold tests* f_a^+ and f_a^-, the characteristic functions of $\{x : a \le x\}$ and of $\{x : a < x\}$, for $a \in 2^{\mathbb{N}}$.

Prefix tests are continuous, and so are $f^+_{t^\frown 0^\infty}$ and $f^-_{t^\frown 1^\infty}$, which depend only on the first $|t|$ bits of x; equality tests, and threshold tests at other points, are discontinuous Baire class 1 functions. All closures below are taken in the topology of pointwise convergence. For a statement S, we write $[\![S]\!]$ for its truth value: 1 if S is true and 0 if it is false.

(1) The *Alexandroff compactification* $A(2^{\mathbb{N}}) = \{\delta_a : a \in 2^{\mathbb{N}}\} \cup \{\mathbf{0}\}$. Each δ_a is isolated, and a neighborhood of $\mathbf{0}$ omits only finitely many δ_a. It is not separable, and not first countable at $\mathbf{0}$.

(2) The *extended Alexandroff compactification* $\hat{A}(2^{\mathbb{N}}) = \overline{\{v_t : t \in 2^{<\mathbb{N}}\}} = \{v_t : t \in 2^{<\mathbb{N}}\} \cup A(2^{\mathbb{N}})$. As $n \to \infty$, the prefix tests $v_{a{\restriction}n}$ converge to δ_a; along an antichain they converge to $\mathbf{0}$. It is separable and not first countable.

(3) The *split Cantor set* $S(2^{\mathbb{N}}) = \{f_a^+, f_a^- : a \in 2^{\mathbb{N}}\}$. The continuous thresholds $f^+_{t^\frown 0^\infty}$, together with the isolated point $\mathbf{0} = f^-_{1^\infty}$, form a countable dense subset: thresholds approaching a from below converge to f_a^+, and from above to f_a^-. It is separable, first countable and hereditarily separable, but not metrizable.

(4) The *Alexandroff duplicate* $D(2^{\mathbb{N}}) = \{(\mathbf{0}, a), (\delta_a, a) : a \in 2^{\mathbb{N}}\} \subseteq A(2^{\mathbb{N}}) \times 2^{\mathbb{N}}$. The points (δ_a, a) are isolated, and a neighborhood of $(\mathbf{0}, a)$ contains $(\mathbf{0}, b)$ and (δ_b, b) for all b near a with finitely many exceptions. It is first countable but not separable. Its separable extension is $\hat{D}(2^{\mathbb{N}}) = \overline{\{(v_t, t^\frown 0^\infty) : t \in 2^{<\mathbb{N}}\}} = D(2^{\mathbb{N}}) \cup \{(v_t, t^\frown 0^\infty) : t \in 2^{<\mathbb{N}}\}$.

(5) The *extended duplicate of the split Cantor set*

$$\hat{D}(S(2^{\mathbb{N}})) = \overline{\{(v_t, f^+_{t^\frown 0^\infty})\}_t} \subseteq \hat{A}(2^{\mathbb{N}}) \times S(2^{\mathbb{N}}),$$

which consists of the points $(v_t, f^+_{t^\frown 0^\infty})$ and, for $a \in 2^{\mathbb{N}}$, the points (δ_a, f^+_a), $(\mathbf{0}, f^+_a)$ and $(\mathbf{0}, f^-_a)$ [ADK08, 4.3.7]. Along the words $t = a{\restriction}n$, the thresholds $t^\frown 0^\infty$ increase to a, which gives the points (δ_a, f^+_a). It is separable and first countable, but the points (δ_a, f^+_a) form an uncountable discrete subspace, so it is not hereditarily separable.

The same constructions apply with $2^{\mathbb{N}}$ replaced by another compact metric space K; we write $A(K)$, $D(K)$, and so on. A compactum of size continuum, such as $A([0,1])$, is homeomorphic to $A(2^{\mathbb{N}})$.

1.5. **NIP as a universal dividing line between polynomial and exponential complexity.** The particular case of the BFT dichotomy (Theorem 1.4) in which A consists of $\{0,1\}$-valued (i.e., yes/no-valued) functions was discovered independently around 1971–1972 in three foundational contexts, each separating polynomial ("tame") from exponential ("wild") complexity: in model theory, by Saharon Shelah [She71, She90]; in combinatorics, by Norbert Sauer [Sau72] and by Shelah [She72, She90]; and in statistical learning, by Vladimir Vapnik and Alexey Chervonenkis [VC71, VC74].

In model theory: Shelah's classification theory is a foundational program in mathematical logic devised to categorize first-order theories based on the complexity and structure of their models. A theory T is considered classifiable in Shelah's sense if the number of non-isomorphic models of T of a given cardinality can be described by a bounded number of numerical invariants. In contrast, a theory T is unclassifiable if the number of models of T of a given cardinality is the maximum possible number. A key fact is that the number of models of T is directly impacted by the number of *types* over sets of parameters in models of T; a controlled number of types is a characteristic of a classifiable theory.

In Shelah's classification program [She90], theories *without* the independence property (called NIP theories, or *dependent* theories) have a well-behaved, "tame" structure: the number of types over a set of parameters of size κ grows slowly (polynomially, or nearly so) in κ.

In contrast, theories with the Independence Property (called IP theories) are considered "intractable" or "wild". A theory with the Independence Property produces the maximum possible number of types over a set of parameters; for a set of parameters of cardinality κ, the theory will have 2^{2^κ}-many distinct types.

In combinatorics: Sauer [Sau72] and Shelah [She72] proved the following independently: Let $\mathscr{F} = \{S_i : i \in \mathbb{N}\}$ be a countable collection of subsets of a set S. Either: for every $n \in \mathbb{N}$ there is a set $A \subseteq S$ with $|A| = n$ such that $|\{S_i \cap A : i \in \mathbb{N}\}| = 2^n$ ($\mathscr{F}$ is a collection having "exponential complexity"); or: there exists $N \in \mathbb{N}$ such that for every $A \subseteq S$ with $|A| \geq N$, one has

$$|\{S_i \cap A : i \in \mathbb{N}\}| \leq \sum_{i=0}^{N-1} \binom{|A|}{i}.$$

($\mathscr{F}$ is a collection having "polynomial complexity"). Clearly, any collection $\mathscr{F}$ of subsets of a *finite* set S has polynomial complexity. The "polynomial" name is justified: indeed, for fixed $N > 0$, as a function of the size $|A| = m > 0$, we have

$$\sum_{i=0}^{N-1} \binom{m}{i} \leq \sum_{i=0}^{N-1} \frac{m^i}{i!} \leq \left(\sum_{i=0}^{N-1} \frac{1}{i!} \right) \cdot m^{N-1} < e \cdot m^{N-1},$$

which is polynomial in m for fixed N.

In machine learning: Readers familiar with statistical learning may recognize the Sauer–Shelah lemma as the dichotomy discovered and proved slightly earlier (1971) by Vapnik and Chervonenkis [VC71, VC74] to address uniform convergence in statistics. The least integer N given by the preceding paragraph, when it exists, is called the *VC-dimension* of $\mathscr{F}$; it is a core concept in machine learning. If such an integer N does not exist, we say that the VC-dimension of $\mathscr{F}$ is infinite. The lemma provides upper bounds on the number of data points (sample size) needed to learn a concept class of known VC dimension d up to a given admissible error in the statistical sense. The Fundamental Theorem of Statistical Learning states that a hypothesis class is PAC-learnable (PAC stands for "Probably Approximately Correct") if and only if its VC dimension is finite.

Historical remarks. The Baire hierarchy of functions was introduced by René-Louis Baire in his 1899 doctoral thesis, *Sur les fonctions de variables réelles*. The field of C_p-theory was pioneered by A. V. Arhangel'skiĭ and his students in the 1970s and 1980s [Ark92], and it has found many applications in model theory and functional analysis. For a recent survey, see [Tka11].

2. Compositional Computation Structures: A structural approach to arbitrary-precision computations

In this section, we connect function spaces with arbitrary-precision computations. We start by summarizing some basic concepts from [ADIW26].

2.1. **Computation States Structures.** A *computation states structure* is a pair $(L, \mathcal{P})$, where L is a set whose elements we call *states* and $\mathcal{P}$ is a collection of real-valued functions on L that we call *predicates*, also called *features*. For each $P \in \mathcal{P}$, we call the value $P(v)$ the P-th *(observable) feature* of v.

A *transition* of a computation states structure $(L, \mathcal{P})$ is a map $f : L \to L$.

The *type* of a state $v \in L$ is the indexed family[3]

$$\mathrm{tp}(v) = (P(v))_{P \in \mathcal{P}} \in \mathbb{R}^{\mathcal{P}}$$

[3]As in Proposition 1.8 above, we regard each predicate on L as named by a specific formal label P (a *predicate symbol*) which is given the semantic meaning of a *bona fide* real function $P(\cdot) : v \mapsto P(v)$ (the predicate proper) in the computation states structure. Such a formal symbol P becomes *de facto* the name of an observable feature having the real value $P(v)$ at any state $v \in L$. Whenever $\mathcal{P}$ is used as an index set (e.g., as in the topological product $\mathbb{R}^{\mathcal{P}}$ below), it is regarded as a set of formal symbols. This syntactic vs. semantic abuse of nomenclature is commonplace in mathematics, but the model-theoretic framework requires contextual awareness of the distinction. To disambiguate when necessary, we shall write $P(\cdot)$, implying the semantic meaning of a real function on L, in contrast to P, meant as a purely syntactic symbol. Nevertheless, by a standard abuse of nomenclature in appropriate contexts, we may also treat P as a function on states (i.e., take P to mean $P(\cdot)$ proper).

of its features.

Intuitively, L is the set of states of a computation, and the predicates $P \in \mathcal{P}$ are primitives that are given and accepted as computable. (I.e., each symbol $P \in \mathcal{P}$ denotes a function $P(\cdot)$ taken as a computational primitive on L.)

We assume that each state $v \in L$ is uniquely characterized by its type $\mathrm{tp}(v)$, so we may identify L with a topological subspace of $\mathbb{R}^{\mathcal{P}}$ and topologized as such (i.e., L is identified with its forward image $\mathrm{tp}[L] \subseteq \mathbb{R}^{\mathcal{P}}$ under the type map). As a topological space, L is thus endowed with the topology of pointwise convergence induced by the product topology of $\mathbb{R}^{\mathcal{P}}$. In particular, for each $P \in \mathcal{P}$, the projection map $v \mapsto P(v)$ is continuous.

Important state spaces include $L = \mathbb{R}^{\mathbb{N}}$ (resp., $L = \mathbb{R}^n$ for some positive integer n), endowed with one predicate $P_i(v) = v_i$ (giving the i-th coordinate of v) for each $i \in \mathbb{N}$ (resp., for each $0 \leq i < n$). State subspaces $L \subseteq \mathbb{R}^{\mathbb{N}}$ and $L \subseteq \mathbb{R}^n$ are of primary interest in this paper. (Any computation states structure having a finite or countable family of predicates is effectively such a subspace.)

Definition 2.1. Given a computation states structure $(L, \mathcal{P})$, any type $\mathrm{tp}(v) \in \mathbb{R}^{\mathcal{P}}$ of a state $v \in L$ will be called a *realized* (state) type. The topological closure of the set of realized types in $\mathbb{R}^{\mathcal{P}}$ (endowed with the pointwise convergence topology) will be called the *space of (state) types* of $(L, \mathcal{P})$, denoted $\mathcal{L}$; elements $\mathbf{v} \in \mathcal{L}$ are called *state types*. Elements $\mathbf{v} \in \mathcal{L} \setminus L$ are *unrealized* state types.

Intuitively, state types capture a notion of "limit state". For each $P \in \mathcal{P}$, the P-th coordinate map (natural projection) $\mathbb{R}^{\mathcal{P}} \to \mathbb{R} : \mathbf{v} = (\mathbf{v}_Q)_{Q \in \mathcal{P}} \mapsto \mathbf{v}_P$, gives a continuous extension to $\mathcal{L}$ of the predicate $P(\cdot)$ on L; abusing notation, the extension will still be denoted $P(\cdot)$ (or just P).

Remark 2.2. One may regard unrealized state types as "hyper-states"; when such a type is obtained as a sequential limit (or ultralimit) of realized state types, it may be called a "deep state". A comment for readers versed in model theory: at least for the purpose of realizing state types (with parameters in the original states space L), the type space $(\mathcal{L}, \mathcal{P})$ serves as a "poor man's" ultraproduct of $(L, \mathcal{P})$ wherein, by the very definition of $\mathcal{L}$, each type $\mathbf{v}$ is realized.

2.1.1. *Bounds on precision and magnitude. Shards.* In order for computation states structures to serve their purpose as models of real-world computing, we must keep in mind that computations involving real-valued quantities can only ever be carried out approximately, to a preset degree of precision (i.e., to within acceptable error bounds). More subtly, yet equally important in practice, one must also bound the magnitude of real quantities involved in all computations: one cannot possibly store—let alone compute with—arbitrarily large numbers! Thus, although the collection $\mathcal{P}$ of predicates may be infinite (possibly, uncountable even), actual computations involve choices, made at the outset, of bounds for errors and for absolute magnitudes of quantities involved. Heuristically, different parts of a computation may require suitably matching (different) error bounds, as well as dealing with quantities of very different sizes. Focusing for now on the second of these issues, our results shall generally apply to sets of computations for which arbitrary but fixed bounds for each predicate are set in advance. This leads to the concept of *shard* first introduced in [ADIW26], as per the following definition.

Definition 2.3. Fix a computation states structure $\underline{L} = (L, \mathcal{P})$, with $L \subseteq \mathbb{R}^{\mathcal{P}}$ implicitly identified with its type space. A *sizer* is a family $r_\bullet = (r_P)_{P \in \mathcal{P}}$ of

positive real numbers, indexed by $\mathcal{P}$. Given a sizer $r_\bullet$, let $\mathbb{R}[r_\bullet] \coloneqq \prod_{P\in\mathcal{P}}[-r_P, r_P]$ (a compact space), and let the $r_\bullet$-*shard* of the states space L be

$$L[r_\bullet] = L \cap \mathbb{R}[r_\bullet].$$

For a sizer $r_\bullet$, the $r_\bullet$-*type shard* is defined as $\mathcal{L}[r_\bullet] = \overline{L[r_\bullet]}$ (a closed, hence compact subset of $\mathbb{R}[r_\bullet]$).

Let also $\mathcal{L}_{\mathrm{sh}}$ (the space of *shard-supported types*) be the union of all type-shards as the sizer $r_\bullet$ varies.

Evidently, $\mathcal{L}_{\mathrm{sh}} \subseteq \mathcal{L}$, although the inclusion may be proper. However, the equality $\mathcal{L}_{\mathrm{sh}} = \mathcal{L}$ holds in the important special case when $\mathcal{P}$ is (at most) countable (see [ADIW26]).

2.2. **Compositional Computation Structures.**

Definition 2.4. A *compositional computation structure* (CCS) is a triple $(L, \mathcal{P}, \Gamma)$, where

- $(L, \mathcal{P})$ is a computation states structure, and
- $\Gamma \subseteq L^L$ is a semigroup under composition.

Elements of the semigroup Γ are called the *computations* of the structure $(L, \mathcal{P}, \Gamma)$. The state space L is implicitly identified with a subset of the type space $\mathcal{L} \subseteq \mathbb{R}^{\mathcal{P}}$ of $(L, \mathcal{P})$. We also assume that the identity map id on L is an element of Γ (which is thus not merely a semigroup but a monoid of transformations of L).

We topologize Γ as a subset of the topological product L^L, where (as usual) the "exponent" L serves merely as an index set, but the "base" $L \subseteq \mathbb{R}^{\mathcal{P}}$ has the relative product topology (of pointwise convergence). In this manner, the semigroup Γ is identified with a subset of the topological product $\mathcal{L}^L \subseteq (\mathbb{R}^{\mathcal{P}})^L$, which leads to an inclusion $\overline{\Gamma} \subseteq \mathcal{L}^L$. Elements $\xi \in \overline{\Gamma}$ are called (real-valued) *deep computations* or *ultracomputations*.

A collection R of sizers is *exhaustive* if $L = \bigcup_{r_\bullet\in R} L[r_\bullet]$ (shards $L[r_\bullet]$ exhaust L). A transformation $\gamma \in \Gamma$ is R-*confined* if γ restricts to a map $\gamma|_{L[r_\bullet]} : L[r_\bullet] \to L[r_\bullet]$ (into $L[r_\bullet]$ itself) for every $r_\bullet \in R$. A subset $\Delta \subseteq \Gamma$ is R-*confined* if each $\gamma \in \Delta$ is.

Unlike its subspace L^L (a topological semigroup), the space $\mathcal{L}^L$ does not have a natural semigroup structure, so ultracomputations are not generally composable. Intuitively, ultracomputations are "final" or "ultimate".

Proposition 2.5. *If $\Delta \subseteq \Gamma$ is confined by an exhaustive sizer collection, then $\overline{\Delta}$ is a compact subset of $\mathcal{L}^L_{\mathrm{sh}}$.*

Proof. If R confines Δ, choose for each $v \in L$ some $r_\bullet^{(v)} \in R$ with $v \in L[r_\bullet^{(v)}]$. Then Δ lies in the compact space $\prod_{v\in L} \mathcal{L}[r_\bullet^{(v)}] \subseteq \mathcal{L}^L_{\mathrm{sh}}$, which therefore contains $\overline{\Delta}$. □

For a CCS $(L, \mathcal{P}, \Gamma)$, we regard the elements of Γ as "standard" finitary computations, and the elements of $\overline{\Gamma}$, i.e., deep computations, as possibly infinitary limits of standard computations.

Remarks 2.6. (For readers versed in model theory.)[4]

[4]Refer to [Kei23] where a general framework for real-valued structures is introduced; see also [DnI17, section 6] for a basic tutorial.

(1) Compositional computation structures as defined above are not real-valued structures in a literal sense, because there is no intrinsic structural manner to ensure that Γ is realized as a literal subset of the product space L^L. However, in a semantically transparent manner, any CCS as above may alternatively be seen (per its original definition in [ADIW26]) as a real-valued structure $(L, \mathcal{P}, \Gamma, \circ, \mathrm{ev})$, where
   - $(L, \mathcal{P})$ is a computation states structure,
   - $(\Gamma, \circ)$ is a semigroup,
   - $\mathrm{ev} : \Gamma \times L \to L : (\gamma, v) \mapsto \mathrm{ev}(\gamma, v)$ is a semigroup action of Γ on L (i.e., $\mathrm{ev}(\gamma \circ \delta, v) = \mathrm{ev}(\gamma, \mathrm{ev}(\delta, v))$ for $\gamma, \delta \in \Gamma$ and $v \in L$).

   Both definitions are reconciled upon identifying $\gamma \in \Gamma$ with the map $(\gamma, v) \mapsto \mathrm{ev}(\gamma, v)$.

(2) In the preceding sense of real-valued structures, the class of compositional computation structures (for a fixed predicate symbol collection $\mathcal{P}$) is first-order elementary. The limited notion of state type we have introduced (essentially, quantifier-free state types with parameters of sort L only) may be enriched to state types involving more formulas (e.g., with parameters in Γ, or including quantified formulas), and extended to "transform-types" (i.e., types of sort Γ). In particular, given a structure $\underline{C} = (L, \mathcal{P}, \Gamma, \circ, \mathrm{ev})$, all types are realized in some ultrapower $\underline{C}^{(\mathcal{U})}$ of $\underline{C}$ (constructed in the usual manner detailed in the references above).[5] This approach to spaces of types was introduced in Banach space theory by Krivine [Kri74, KM81, Kri72].

2.3. **Computability of deep computations and the Extendibility Axiom.** Let $(L, \mathcal{P})$ be a computation states structure with *countable* predicate collection $\mathcal{P}$.

(1) A *computable predicate* is any real function $\varphi : L \to \mathbb{R}$ whose restriction to an arbitrary shard is uniformly approximable by polynomials (with real coefficients) in the features $P(\cdot)$.
(2) A transform $f : L \to \mathcal{L}$ is *computable* if, for each $Q \in \mathcal{P}$, the output feature $Q \circ f : L \to \mathbb{R}$ is a computable predicate.

It is shown in [ADIW26] that a transform $f : L \to \mathcal{L}$ is computable, that is, has uniformly polynomially approximable features on shards, if and only if it extends to a continuous map $\tilde{f} : \mathcal{L} \to \mathcal{L}$. This motivates the following definition.

A computation $\gamma \in \Gamma$ is *extendable* if it has some extension $\tilde{\gamma} : \mathcal{L}_{\mathrm{sh}} \to \mathcal{L}_{\mathrm{sh}}$ such that, for every sizer $r_\bullet$, there is a sizer $s_\bullet$ such that $\tilde{\gamma}|_{\mathcal{L}[r_\bullet]} : \mathcal{L}[r_\bullet] \to \mathcal{L}[s_\bullet]$ is continuous.

Such an extension is unique if it exists, since it is continuous on each type shard and $L[r_\bullet]$ is dense in $\mathcal{L}[r_\bullet]$.

Abusing notation, for any $\Delta \subseteq \Gamma$ consisting of extendable computations, the set $\{\tilde{\gamma} \colon \gamma \in \Delta\}$ will be denoted $\tilde{\Delta}$.

Extendibility Axiom. *Every computation $\gamma \in \Gamma$ is extendable.*

For a more detailed discussion of the Extendibility Axiom, we refer the reader to [ADIW26].

For the rest of the paper, the Extendibility Axiom is imposed on all CCSs.

The following lemma extends deep computations to the type shards; it is used in Theorems 3.1 and 6.7.

[5] By contrast (cf., Remark 2.2), the ultrapower construction is not necessary to realize state types in a computation states structure.

Lemma 2.7. *Let R be an exhaustive collection of sizers and let $\Delta \subseteq \Gamma$ be R-confined. Every deep computation $f \in \overline{\Delta}$ has an extension $\tilde{f} : \mathcal{L}_{\mathrm{sh}} \to \mathcal{L}_{\mathrm{sh}}$ such that, for every $r_\bullet \in R$, $\tilde{f}$ maps $\mathcal{L}[r_\bullet]$ into itself and $\tilde{f}|_{\mathcal{L}[r_\bullet]} \in \overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$.*

Proof. Write $f = \mathcal{U}\lim_i \gamma_i$ with $\gamma_i \in \Delta$ and $\mathcal{U}$ an ultrafilter. Each $v \in \mathcal{L}_{\mathrm{sh}}$ lies in some type shard $\mathcal{L}[r_\bullet]$, and by the Extendibility Axiom all the $\tilde{\gamma}_i(v)$ lie in a compact type shard $\mathcal{L}[s_\bullet]$, so $\tilde{f}(v) \coloneqq \mathcal{U}\lim_i \tilde{\gamma}_i(v)$ exists, and $\tilde{f}$ extends f. If $r_\bullet \in R$, each $\tilde{\gamma}_i$ maps $L[r_\bullet]$ into itself, hence, by continuity, its closure $\mathcal{L}[r_\bullet]$ into itself; so does $\tilde{f}$, and $\tilde{f}|_{\mathcal{L}[r_\bullet]} = \mathcal{U}\lim_i \tilde{\gamma}_i|_{\mathcal{L}[r_\bullet]}$. □

2.4. **Stability.** The computable maps are the continuous ones. At the top of the hierarchy of this paper sit the sets of computations all of whose deep computations are computable.

Definition 2.8. A set $\Delta \subseteq \Gamma$ of computations is *stable* if, for every sizer $r_\bullet$, every deep computation in $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$ is continuous, that is, computable.

This is the class studied in [ADIW26], where it is characterized by Grothendieck's double limit criterion and identified with stability in the sense of model theory.

2.5. **Two motivating examples.** Before developing the classification, we describe two computations whose deep computations behave very differently. Both are treated in detail in Section 4.

Square roots. Newton's method for $p(x) = x^2 - a$, with $a > 0$, acts on the real projective line $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$ by $N_p(x) = (x^2 + a)/(2x)$. Its iterates are continuous, and they converge pointwise to the map $\mathfrak{f}$ that sends every positive number to $\sqrt{a}$, every negative number to $-\sqrt{a}$, and 0 and ∞ to ∞ (Figure 1). The deep computations are the iterates together with $\mathfrak{f}$. The map $\mathfrak{f}$ is discontinuous, so it is not computable and the iterates are not stable; but it is the limit of a *sequence* of computations, and the space of deep computations is countable.

Newton's method on the Riemann sphere. For $\rho(z) = z^3 - 2z + 2$, Newton's method on the Riemann sphere $\hat{\mathbb{C}}$ behaves in three ways at once (Figure 2 in Section 4.1). On the basins of the three roots the iterates converge. On the basin of the cycle $0 \mapsto 1 \mapsto 0$ they do not: the even and the odd iterates converge to two different deep computations. On the common boundary of these basins, a fractal set, the iterates are chaotic, and for the simpler polynomial $z^2 - 1$ we shall see that some deep computations are not even limits of sequences of computations (Proposition 5.10).

The classification of Section 3 is designed to separate such behaviors. Theorem 3.1 characterizes the sets of computations whose deep computations are limits of sequences, as for square roots, and Todorčević's trichotomy measures how complicated the space of such deep computations can be.

3. Classifying deep computations

3.1. **NIP, Rosenthal compacta, and deep computations.** This is the main section of the paper. We use topology of function spaces to obtain a new hierarchy of complexity classes for deep computations. The main topological tool is Todorčević's trichotomy for Rosenthal compacta [Tod99]. Our inspiration is the work of Glasner and Megrelishvili on tame semigroups in topological dynamics [GM22].

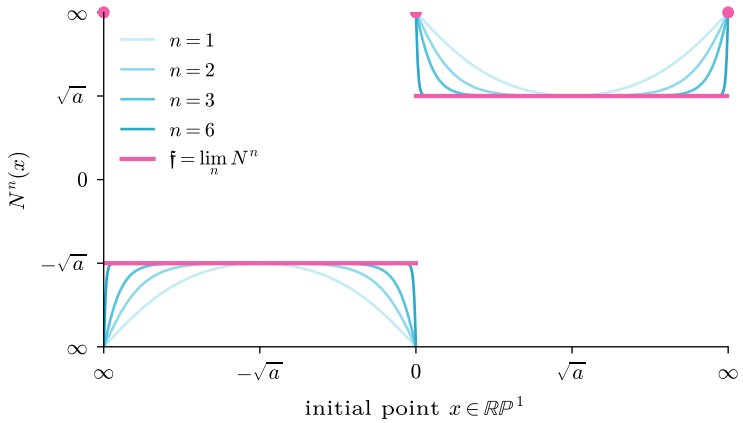


FIGURE 1. Newton iterates N_p^n for $p(x) = x^2 - a$ on the real projective line, drawn on the circle with ∞ at both ends of each axis. The continuous maps N_p^n converge pointwise to the deep computation $\mathfrak{f}$, which sends the positive half line to $\sqrt{a}$, the negative half line to $-\sqrt{a}$, and 0 and ∞ to ∞.

Theorem 3.1, the cornerstone of the section, extends the results of [ADIW26] from continuous deep computations to deep computations of the first Baire class.

Theorem 3.1. *Let $(L, \mathcal{P}, \Gamma)$ be a compositional computation structure (Definition 2.4) satisfying the Extendibility Axiom, with $\mathcal{P}$ countable. Let R be an exhaustive collection of sizers, and suppose that $\Delta \subseteq \Gamma$ is R-confined. Then the following conditions are equivalent:*

(i) $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}} \subseteq \mathrm{B}_1(\mathcal{L}[r_\bullet], \mathcal{L}[r_\bullet])$ *for all* $r_\bullet \in R$*;*

(ii) $\pi_P \circ \Delta|_{L[r_\bullet]}$ *satisfies the NIP for all* $P \in \mathcal{P}$ *and* $r_\bullet \in R$*; that is, for all* $P \in \mathcal{P}$*,* $r_\bullet \in R$*,* $a < b$*,* $\{\gamma_n\}_{n\in\mathbb{N}} \subseteq \Delta$ *there are finite disjoint* $E, F \subseteq \mathbb{N}$ *such that*

$$L[r_\bullet] \cap \bigcap_{n\in E} (\pi_P \circ \gamma_n)^{-1}(-\infty, a] \cap \bigcap_{n\in F} (\pi_P \circ \gamma_n)^{-1}[b, \infty) = \emptyset.$$

Furthermore, if any (hence all) of the preceding conditions hold, then every deep computation $f \in \overline{\Delta}$ can be extended to a Baire-1 function on shards: there is $\tilde{f} : \mathcal{L}_{sh} \to \mathcal{L}_{sh}$ such that $\tilde{f}|_{\mathcal{L}[r_\bullet]} \in \mathrm{B}_1(\mathcal{L}[r_\bullet], \mathcal{L}[r_\bullet])$ for all $r_\bullet \in R$. In particular, on each shard, every deep computation is the pointwise limit of a countable sequence of computations.

Proof. Since $\mathcal{P}$ is countable, $\mathcal{L}[r_\bullet] \subseteq \mathbb{R}^{\mathcal{P}}$ is Polish. The Extendibility Axiom implies that $\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}$ is a pointwise bounded set of continuous functions for all $P \in \mathcal{P}$. Hence, Theorem 1.9 and Lemma 1.10 prove the equivalence of (i) and (ii). If (i) holds, the extension $\tilde{f}$ of Lemma 2.7 satisfies $\tilde{f}|_{\mathcal{L}[r_\bullet]} \in \overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}} \subseteq \mathrm{B}_1(\mathcal{L}[r_\bullet], \mathcal{L}[r_\bullet])$ for all $r_\bullet \in R$. The assertion that every deep computation is a pointwise limit of a countable sequence of computations follows from the fact that for a Polish space X, every compact subset of $\mathrm{B}_1(X)$ is Fréchet–Urysohn (that is, a space where any point in the closure of a set A is the limit of a sequence of points in A; see Theorem 3F in [BFT78] or Theorem 4.1 in [Deb13]). □

3.2. **Todorčević's trichotomy and levels of NIP.** Recall from Section 1.4 the notions of hereditarily separable and first countable spaces. Every separable metrizable space is hereditarily separable, and R. Pol proved that every hereditarily separable Rosenthal compactum is first countable (see section 10 of [Deb13]). This suggests the following definition:

Definition 3.2. Let $(L, \mathcal{P}, \Gamma)$ be a CCS satisfying the Extendibility Axiom and R be an exhaustive collection of sizers. Let $\Delta \subseteq \Gamma$ be an R-confined countable set of computations satisfying the NIP on shards and features (condition (ii) in Theorem 3.1). We say that Δ is:

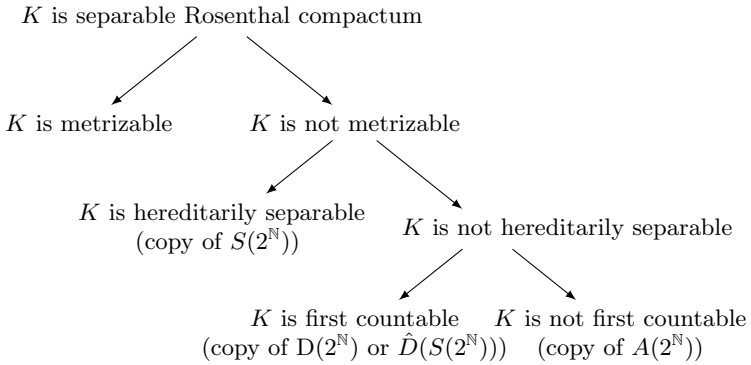
(i) NIP_1 if $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$ is first countable for every $r_\bullet \in R$;
(ii) NIP_2 if $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$ is hereditarily separable for every $r_\bullet \in R$;
(iii) NIP_3 if $\overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$ is metrizable for every $r_\bullet \in R$.

Observe that $\mathrm{NIP}_3 \Rightarrow \mathrm{NIP}_2 \Rightarrow \mathrm{NIP}_1 \Rightarrow \mathrm{NIP}$. Moreover, a countable stable set of computations (Definition 2.8) is NIP_3: on each shard its deep computations form a separable compact subset of a countable product of spaces $\mathrm{C_p}(\mathcal{L}[r_\bullet])$, that is, a separable Eberlein compactum, and separable Eberlein compacta are metrizable [Ark92]. The five compacta of Section 1.4 show that the converse implications fail for Rosenthal compacta: $\hat{A}(2^{\mathbb{N}})$ is separable but not first countable; $\hat{D}(S(2^{\mathbb{N}}))$ and $\hat{D}(2^{\mathbb{N}})$ are separable and first countable but not hereditarily separable; and $S(2^{\mathbb{N}})$ is hereditarily separable but not metrizable. Todorčević proved that these examples are canonical [Tod99].

Theorem 3.3 (Todorčević's Trichotomy, [Tod99], Theorem 3 in [ADK08])**.** *Let K be a separable Rosenthal compactum.*

(i) If K is hereditarily separable but non-metrizable, then $S(2^{\mathbb{N}})$ embeds into K.
(ii) If K is first countable but not hereditarily separable, then either $\mathrm{D}(2^{\mathbb{N}})$ or $\hat{D}(S(2^{\mathbb{N}}))$ embeds into K.
(iii) If K is not first countable, then $A(2^{\mathbb{N}})$ embeds into K.

We thus have the following classification for separable Rosenthal compacta:



The three levels can also be read without topology, as statements about how many inputs are needed to tell deep computations apart. Fix $r_\bullet \in R$ and write $K_{r_\bullet} = \overline{\tilde{\Delta}|_{\mathcal{L}[r_\bullet]}}$. Identify each $f \in K_{r_\bullet}$ with the real-valued function $(P, x) \mapsto P(f(x))$ on $X_{r_\bullet} = \mathcal{P} \times \mathcal{L}[r_\bullet]$, whose points we call *tests*: a test applies the feature P to the output on the input x. The topology of pointwise convergence on $K_{r_\bullet}$ is

then the topology of pointwise convergence on tests. A set D of tests *separates* a set of deep computations if any two of them differ on some test in D.

Proposition 3.4. *Let Δ be as in Definition 3.2 and $r_\bullet \in R$.*

(i) $K_{r_\bullet}$ is metrizable if and only if one countable set of tests separates $K_{r_\bullet}$.
(ii) $K_{r_\bullet}$ is first countable if and only if every $f \in K_{r_\bullet}$ is countably determined*: some countable set D_f of tests separates f from every other member of $K_{r_\bullet}$.*
(iii) $K_{r_\bullet}$ is hereditarily separable if and only if there is no set $E \subseteq K_{r_\bullet}$ of cardinality $2^{\aleph_0}$ whose members are finitely separated*: for each $f \in E$ there are finitely many tests and $\varepsilon_f > 0$ such that every $g \in E \setminus \{f\}$ differs from f by at least ε_f on one of them.*

Proof. (i) If a countable set D of tests separates $K_{r_\bullet}$, restriction to D is a continuous injection of the compact space $K_{r_\bullet}$ into the metrizable space $\mathbb{R}^D$, hence a homeomorphism onto its image. Conversely, if $K_{r_\bullet}$ is metrizable, then $K_{r_\bullet} \times K_{r_\bullet}$ is second countable, so the open set $\{(f,g) : f \neq g\}$ is Lindelöf; it is covered by the open sets $\{(f,g) : f(t) \neq g(t)\}$, for tests t, and a countable subcover gives the required set D. (ii) If f is countably determined, $\{f\}$ is a countable intersection of open sets, and in a compact Hausdorff space such a point has a countable base of neighborhoods. Conversely, if (U_k) is a countable base at f and each U_k contains a basic neighborhood defined by a finite set F_k of tests, then $D_f = \bigcup_k F_k$ works. (iii) A set is finitely separated exactly when it is a discrete subspace, and, by Theorem 3.1, $K_{r_\bullet}$ is a separable Rosenthal compactum; for such compacta, hereditary separability is equivalent to the absence of discrete subspaces of cardinality $2^{\aleph_0}$, by a theorem of Todorčević [Tod99] (see also [GM22, Theorem 0.4]). □

Thus Δ is NIP_3 when one countable set of tests decides every deep computation, NIP_1 when each deep computation is decided by countably many tests of its own, and NIP_2 when no continuum of deep computations can be told apart one by one with finitely many tests.

Section 5 gives computations at each level of the hierarchy, and shows that every implication between the classes stable, NIP_3, NIP_2, NIP_1 and NIP is strict.

The definitions of NIP_1, NIP_2 and NIP_3 given here are topological; whether they admit a non-topological characterization is Question 7.1.

3.3. **The Argyros–Dodos–Kanellopoulos heptachotomy and binary search.** In the separable compacta of Section 1.4, the countable dense subsets are indexed by the binary tree $2^{<\mathbb{N}}$. This indexing suits our computational emphasis: real numbers are represented by infinite binary sequences, i.e., branches of the binary tree, and approximated by finite bitstrings, i.e., its nodes.

Definition 3.5. Let X be a Polish space.

(1) If I is countable and $\{f_i : i \in I\} \subseteq \mathbb{R}^X$, $\{g_i : i \in I\} \subseteq \mathbb{R}^X$ are two families indexed by I, we say that they are *equivalent* if the map $f_i \mapsto g_i$ extends to a homeomorphism from $\overline{\{f_i : i \in I\}}$ to $\overline{\{g_i : i \in I\}}$.
(2) If $\{f_t : t \in 2^{<\mathbb{N}}\}$ is a pointwise bounded family, we say that it is *minimal* if for every dyadic subtree $\{s_t : t \in 2^{<\mathbb{N}}\}$ of $2^{<\mathbb{N}}$, $\{f_{s_t} : t \in 2^{<\mathbb{N}}\}$ is equivalent to $\{f_t : t \in 2^{<\mathbb{N}}\}$.

One of the main results in [ADK08] is that, up to equivalence, there are seven minimal families and that for every relatively compact $\{f_t : t \in 2^{<\mathbb{N}}\} \subseteq \mathrm{B}_1(X)$ there

is a dyadic subtree $\{s_t : t \in 2^{<\mathbb{N}}\}$ such that $\{f_{s_t} : t \in 2^{<\mathbb{N}}\}$ is equivalent to one of the minimal families. We shall describe the seven minimal families next. We follow the notation of [ADK08]. Fix a regular dyadic subtree $T = \{s_t : t \in 2^{<\mathbb{N}}\}$ of $2^{<\mathbb{N}}$ (i.e., a dyadic subtree such that every level of T is contained in a level of $2^{<\mathbb{N}}$) with the property that for all $s, s' \in T$, we have $s^\frown 0^\infty \neq s'^\frown 0^\infty$ and $s^\frown 1^\infty \neq s'^\frown 1^\infty$. We use the prefix tests v_t and the threshold tests $f_a^\pm$ of Section 1.4. A pair (f, g) of functions on $2^{\mathbb{N}}$ is regarded as a single function on the disjoint union $2^{\mathbb{N}} \sqcup 2^{\mathbb{N}}$.

(1) $D_1 = \{\frac{1}{|t|+1} v_t : t \in 2^{<\mathbb{N}}\}$. Its closure is $D_1 \cup \{\mathbf{0}\}$, the one-point compactification of a countable discrete set.
(2) $D_2 = \{s_t^\frown 0^\infty : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_2} = 2^{\leq\mathbb{N}}$.
(3) $D_3 = \{f^+_{s_t^\frown 0^\infty} : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_3} = S(2^{\mathbb{N}})$.
(4) $D_4 = \{f^-_{s_t^\frown 1^\infty} : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_4} = S(2^{\mathbb{N}})$.
(5) $D_5 = \{v_t : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_5} = \hat{A}(2^{\mathbb{N}})$.
(6) $D_6 = \{(v_{s_t}, s_t^\frown 0^\infty) : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_6} = \hat{D}(2^{\mathbb{N}})$.
(7) $D_7 = \{(v_{s_t}, f^+_{s_t^\frown 0^\infty}) : t \in 2^{<\mathbb{N}}\}$. This is discrete in $\overline{D_7} = \hat{D}(S(2^{\mathbb{N}}))$.

Theorem 3.6 (Heptachotomy of minimal families, Theorem 2 in [ADK08])**.** *Let X be Polish. For every relatively compact $\{f_t : t \in 2^{<\mathbb{N}}\} \subseteq \mathrm{B}_1(X)$, there exists $i = 1, 2, \ldots, 7$ and a regular dyadic subtree $\{s_t : t \in 2^{<\mathbb{N}}\}$ of $2^{<\mathbb{N}}$ such that $\{f_{s_t} : t \in 2^{<\mathbb{N}}\}$ is equivalent to D_i. Moreover, the families D_i are minimal and mutually non-equivalent.*

The implication of this result for deep computations is the following: the building blocks of the families $D_1, \ldots, D_7$ are primitives of binary search on $2^{\mathbb{N}}$, the space of binary expansions. A node $t \in 2^{<\mathbb{N}}$ is a dyadic interval $[t^\frown 0^\infty, t^\frown 1^\infty]$, the set of inputs that round to t at precision $|t|$. The prefix test v_t asks whether the input rounds to t; the point $t^\frown 0^\infty$ is the rounded value itself; the comparisons $f^+_{t^\frown 0^\infty}$ and $f^-_{t^\frown 1^\infty}$ ask whether the input lies to the right of the left endpoint, or strictly to the right of the right endpoint, of the interval. Table 1 reads the seven families accordingly.

Family	Function at node t	Limits
D_1	prefix test weighted by $1/(\|t\| + 1)$	the test that never succeeds: confidence vanishes with depth
D_2	return the rounded value $t^\frown 0^\infty$	the exact input: successive approximation
D_3	compare with the left endpoint	comparison with an arbitrary real, ties broken upward
D_4	compare with the right endpoint	comparison with an arbitrary real, ties broken downward
D_5	prefix test	exact equality, or no match
D_6	prefix test and rounded value	a dictionary lookup that returns its key
D_7	prefix test and comparison	a three-way comparison: below, inside or above

TABLE 1. The minimal families of Argyros, Dodos and Kanellopoulos as primitives of binary search.

By the theorem of Argyros, Dodos and Kanellopoulos, if $\{\gamma_t : t \in 2^{<\mathbb{N}}\}$ is an NIP family of computations indexed by the binary tree and P is a feature, then, on each shard, the family $\{P \circ \gamma_t\}$ contains, along a dyadic subtree, a family equivalent to one of these seven.

4. Examples of Compositional Computation Structures

We now turn from the general theory to examples, and compute the deep computations of three compositional computation structures explicitly: Newton's method on the Riemann sphere for two cubic polynomials (Sections 4.1 and 4.2), and Newton's method for square roots on the real projective line (Section 4.3). Section 5 then places these and other computations at each level of the hierarchy of Section 3.

We begin by formalizing the Newton–Raphson root-finding method as a CCS [Gal00], [BFB16, Section 2.3]. Let $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ be the extended complex plane (the Alexandroff —one-point— compactification of $\mathbb{C}$). We identify $\hat{\mathbb{C}}$ with the unit sphere $S^2 \subseteq \mathbb{R}^3$ (the Riemann sphere) through the inverse stereographic projection $z \mapsto (P_1(z), P_2(z), P_3(z))$, a homeomorphism $\hat{\mathbb{C}} \to S^2$, where $(P_1(\infty), P_2(\infty), P_3(\infty)) = (0, 0, 1)$ and, for $z \neq \infty$,

$$P_1(z) = \frac{2\,\Re z}{|z|^2 + 1}, \qquad P_2(z) = \frac{2\,\Im z}{|z|^2 + 1}, \qquad P_3(z) = \frac{|z|^2 - 1}{|z|^2 + 1}.$$

The extended complex plane $\hat{\mathbb{C}}$ is thus endowed with the real predicates P_1, P_2, P_3 with values in $[-1, 1]$; its topology is metrized by the pullback of the usual Euclidean metric on S^2.

Newton CCSs: steps and fractals. Let $p = p(z)$ be a non-constant rational function with complex coefficients —i.e., an analytic (meromorphic) map $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$. The *Newton step* (or *Newton map*) for p is the rational map

$$N_p : z \mapsto z - \frac{p(z)}{p'(z)}. \tag{4.1}$$

Successive iterations of the Newton–Raphson method (i.e., compositional powers $N_p^{(n)}$) may be regarded as computations of a CCS whose states space is $L = \hat{\mathbb{C}}$. The natural choice of predicates is $\mathcal{P} = (P_1, P_2, P_3)$, the inverse stereographic projection coordinates. (The real- and imaginary part maps $z \mapsto \Re z$, $z \mapsto \Im z$ are not defined at $z = \infty$.)

Let $\Gamma_p := \{N_p^{(n)} : n \in \mathbb{N}\}$ be the semigroup of iterates of the Newton map (4.1). The CCS $(\hat{\mathbb{C}}, (P_1, P_2, P_3), \Gamma_p)$ is *Newton's method for* $p(z)$ (or the *Newton CCS for* $p(z)$). Note that the states space $\hat{\mathbb{C}}$ is compact and the Newton map N_p is continuous thereon, so the Extendibility Axiom is satisfied by Newton CCSs.

Given an initial estimate z_0, one hopes for the sequence of Newton steps applied to z_0 (namely, $N_p^{(\bullet)}(z_0) := (z_n)_{n\in\mathbb{N}}$, where $z_n := N_p^{(n)}(z_0)$) to converge to a root r of p.

Fatou and Julia sets. Recall that the *Fatou set* $\mathcal{F}_p$ consists of all points that have a neighborhood on which every sequence of iterates of p has a uniformly convergent subsequence; the *Julia set* $\mathcal{J}_p := \hat{\mathbb{C}} \setminus \mathcal{F}_p$ is its complement: it is (necessarily closed and) typically a fractal. The Julia set of the Newton map N_p is called the *Newton fractal* of p.

4.1. **Example: The Newton CCS for the cubic $\rho(z) = z^3 - 2z + 2$.** The polynomial $\rho(z) = z^3 - 2z + 2$ has Newton map

$$N_\rho : z \mapsto z - \frac{z^3 - 2z + 2}{3z^2 - 2} = \frac{2z^3 - 2}{3z^2 - 2}.$$

Denote the real root of ρ by r_0, and the roots in the upper and the lower half plane by r_+ and r_-. Starting from $z_0 = 0$, the Newton steps alternate $0, 1, 0, 1, \ldots$, and neither 0 nor 1 is a root of ρ.

Let B_0, B_+, B_- be the basins of the roots r_0, r_+, r_-, and let B_{01} be the basin of the attracting cycle $0, 1, 0, 1, \ldots$. The Fatou set of N_ρ is $B_{01} \cup B_0 \cup B_+ \cup B_-$, and each of these four sets is nonempty and open. Since B_{01} is nonempty, the iterates $N_\rho^{(n)}$ do not converge pointwise; however, the even and the odd iterates converge pointwise on B_{01}, to two distinct deep computations f_0 (constant 0 on B_{01}) and f_1 (constant 1 on B_{01}).[6] Figure 2 shows f_0 and f_1. On the Newton fractal of ρ, the behavior of the iterates is chaotic.

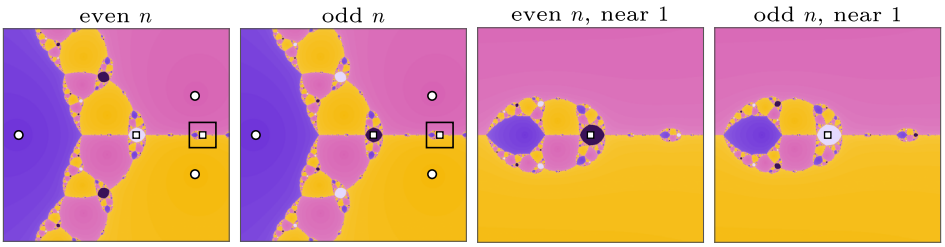


FIGURE 2. The deep computations of Newton's method for $\rho(z) = z^3 - 2z + 2$: the limits of the even iterates $N_\rho^{(2k)}$ and of the odd iterates $N_\rho^{(2k+1)}$, on the whole plane and near the point 1. Points in the basins of the real root and of the roots in the upper and lower half planes (white circles) are purple, orchid and gold, paler where convergence is slower. The remaining open region is the basin of the superattracting cycle $0 \mapsto 1 \mapsto 0$ (white squares). A point there is lilac if the iterate in question is near 0 and plum if it is near 1, so the two colors exchange between even and odd iterates: the two deep computations agree on the basins of the roots and differ on the basin of the cycle. The boundaries of the colored regions form the Newton fractal of ρ.

4.2. **Example: The Newton CCS for the 3-cyclotomic polynomial $\kappa(z) = z^3 - 1$.** To visualize the iterates themselves, rather than their limits, color each z_0 by blending the colors of the roots with relative weights $|z_n - r_i|^{-3}$, where $z_n := N_p^{(n)}(z_0)$. This coloring is a continuous function of z_0, so at first it is blurred; as n increases it sharpens into large regions of nearly constant color, separated by narrow regions where the color varies rapidly, that is, where the coloring is nearly

[6]The point ∞ is a repelling fixed point of N_ρ: if $|z|$ is large, then $N_\rho(z) = \frac{2}{3}z + O(1)$. Hence ∞ lies on the Newton fractal of ρ, and $f_0(\infty) = \infty = f_1(\infty)$.

discontinuous. Figure 3 shows this for $\kappa(z) = z^3 - 1$, whose roots are the cube roots of unity $r_0 = 1$ and $r_\pm = \exp(\pm 2\pi i/3)$, and whose Newton map is

$$N_\kappa : z \mapsto z - \frac{z^3 - 1}{3z^2} = \frac{2z^3 + 1}{3z^2}.$$

Unlike N_ρ, the map N_κ has no attracting cycle: the Fatou set is the union of the three basins, and on each of them the iterates converge to the corresponding root.

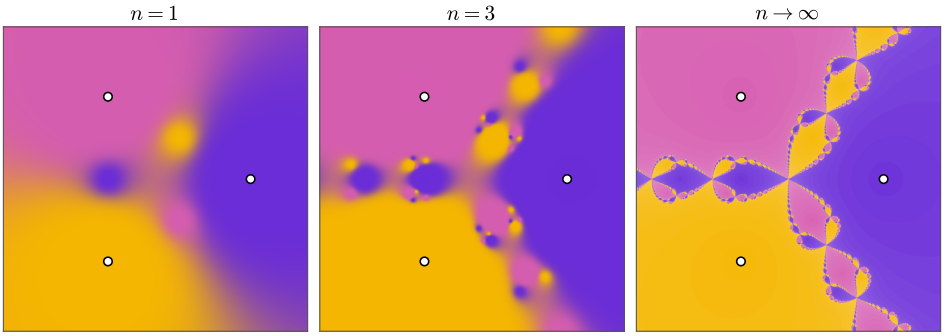


FIGURE 3. Newton's method for $\kappa(z) = z^3 - 1$ after one and three iterations, and its limit. The coloring blends the colors of the three cube roots of unity (white circles) according to the distance of $N_\kappa^{(n)}(z)$ to each of them; the limit is constant on each basin and discontinuous across the Newton fractal; in the right panel, points are paler where convergence is slower.

The chaos of Newton's method on its Julia set is visible to the predicates P_1, P_2, P_3, and it can make a Newton CCS on the Riemann sphere fail to be NIP (Proposition 5.10).

4.3. **Example: Square roots.** When $p : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ has real coefficients, it restricts to a map $\mathbb{RP}^1 \to \mathbb{RP}^1$ on the real-projective line $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$. In this case, the Newton CCS for p may be regarded as having states space $\mathbb{RP}^1$. We shall explain the details in the simplest case of a polynomial $p(x) = x^2 - a$ (real $a > 0$) to illustrate iterative square-root approximations.

Let $\mathbb{R} \to S^1 : x \mapsto (P_1(x), P_2(x))$ be the (inverse) stereographic projection of the real line onto $S^1 \subseteq \mathbb{R}^2$, i.e.,

$$P_1(x) = \frac{2x}{x^2 + 1}, \qquad P_2(x) = \frac{x^2 - 1}{x^2 + 1},$$

extended to a projective transformation of the real projective line $\mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$ by $\infty \mapsto (P_1(\infty), P_2(\infty)) = (0, 1)$; this gives a compact computation states structure $(\mathbb{RP}^1, (P_1, P_2))$. The Newton map N_p is

$$x \mapsto x - \frac{x^2 - a}{2x} = \begin{cases} \frac{x^2 + a}{2x}, & x \neq 0, \infty; \\ \infty, & x = 0, \infty. \end{cases}$$

These Newton steps converge to one of the square roots $\pm\sqrt{a}$ starting from any initial point $x_0 \neq 0, \infty$. More explicitly, the unique deep equilibrium $\mathfrak{f} : \mathbb{RP}^1 \to \mathbb{RP}^1$

of (the sequence of iterates of) N_p is given by

$$\mathfrak{f}(x) = \lim_{n\to\infty} N_p^n(x) = \begin{cases} \sqrt{a}, & x \in (0,\infty); \\ -\sqrt{a}, & x \in (\infty, 0); \\ \infty, & x = 0, \infty. \end{cases}$$

Thus, $\mathfrak{f}$ is discontinuous at $x = 0, \infty$ (Figure 1).

5. Computations across the hierarchy

The hierarchy

$$\text{stable} \Rightarrow \text{NIP}_3 \Rightarrow \text{NIP}_2 \Rightarrow \text{NIP}_1 \Rightarrow \text{NIP}$$

of Section 3 was defined through the topology of the space of deep computations. In this section we see that each of its levels is occupied by a classical algorithm, so that every implication is strict, and that some familiar algorithms lie beyond it. In each example the computations are compositions of the steps of an algorithm. At all levels but one they are the powers of a single step map; at the level NIP but not NIP_1 they form a group generated by two steps, as in the Euclidean algorithm. The level of each algorithm is then a theorem about the closure of its computations, and for such computations the hierarchy coincides with a hierarchy studied in topological dynamics by Glasner and Megrelishvili [GM22]. We begin with this observation. Table 2 at the end of the section summarizes the examples.

5.1. **Iterative algorithms and tame dynamics.** The algorithms of this section act on a compact space of states, and each of their steps is continuous. We isolate this situation. A *dynamical CCS* is a compositional computation structure $(L, \mathcal{P}, \Gamma)$ in which L is a compact metrizable space, $\mathcal{P}$ is a countable family of continuous predicates that separates the points of L, and Γ is a countable semigroup of continuous maps $L \to L$ that contains the identity. It is a *cascade CCS* if $\Gamma = \{\gamma^n : n \in \mathbb{N}\}$ is the semigroup of iterates of a single continuous map $\gamma\colon L \to L$. Since $\mathcal{P}$ separates points and L is compact, the map $v \mapsto \text{tp}(v)$ is a homeomorphism of L onto the space of types, so L is covered by a single shard, every type is realized, and the Extendibility Axiom holds with $\tilde{\gamma} = \gamma$ for every $\gamma \in \Gamma$. The deep computations of a dynamical CCS form the pointwise closure

$$E(L, \Gamma) = \overline{\Gamma} \subseteq L^L,$$

which is the *enveloping semigroup* of the dynamical system (L, Γ) introduced by Ellis. For a cascade CCS we write $E(L, \gamma)$. The system (L, Γ) is called *tame* if, for every continuous $f\colon L \to \mathbb{R}$, the family $\{f \circ \gamma : \gamma \in \Gamma\}$ has the NIP of Definition 1.5 [Köh95, Gla06].

A map $p\colon L \to L$ is *of Baire class* 1 if $f \circ p$ is of Baire class 1 for every continuous $f\colon L \to \mathbb{R}$. It suffices that $P \circ p$ be of Baire class 1 for every $P \in \mathcal{P}$. Indeed, the type map embeds L into $\mathbb{R}^{\mathcal{P}}$, and $\mathcal{P}$ is countable, so every open subset of L is a countable union of finite intersections of sets $P^{-1}(U)$, with $P \in \mathcal{P}$ and $U \subseteq \mathbb{R}$ open; by Fact 1.2, the preimage of every open set under p is then an F_σ set, and so is the preimage of every open set under $f \circ p$.

Proposition 5.1. *A dynamical CCS is NIP if and only if the dynamical system (L, Γ) is tame, and if and only if every deep computation is of Baire class* 1*. In that case it is NIP_1, NIP_2 or NIP_3 if and only if $E(L, \Gamma)$ is, respectively, first countable, hereditarily separable or metrizable.*

Proof. The single sizer $r_\bullet$ with $L = L[r_\bullet]$ forms an exhaustive collection that confines Γ. If (L,Γ) is tame, the CCS is NIP: take $f = P$ for $P \in \mathcal{P}$. If the CCS is NIP, then, by Theorem 3.1 and Proposition 1.8, $P \circ p$ is of Baire class 1 for every $p \in E(L,\Gamma)$ and $P \in \mathcal{P}$, so every deep computation is of Baire class 1. Finally, suppose that every deep computation is of Baire class 1, and let $f\colon L \to \mathbb{R}$ be continuous. The map $p \mapsto f \circ p$ is continuous on L^L, so the closure of $\{f \circ \gamma : \gamma \in \Gamma\}$ is the compact set $\{f \circ p : p \in E(L,\Gamma)\}$, which lies in $\mathrm{B}_1(L)$. By Theorem 1.6, the family $\{f \circ \gamma : \gamma \in \Gamma\}$ has the NIP, so (L,Γ) is tame. The last statement is Definition 3.2, since $\overline{\tilde{\Gamma}|_L} = E(L,\Gamma)$. □

For metrizable systems, Glasner and Megrelishvili call these three classes *tame*$_1$, *tame*$_2$ and *hereditarily non-sensitive*. They prove the inclusions between these classes [GM22, Proposition 0.6], and they show by examples that the inclusions are proper.

The simplest iterative methods are those whose iterates converge. Their level is decided by the continuity of the limit.

Proposition 5.2. *Let $(L,\mathcal{P},\Gamma)$ be a cascade CCS whose iterates γ^n converge pointwise to a map $\phi\colon L \to L$. Then $E(L,\gamma) = \{\gamma^n : n \in \mathbb{N}\} \cup \{\phi\}$, and the CCS is NIP$_3$. It is stable if and only if ϕ is continuous.*

Proof. A convergent sequence, together with its limit, is a compact subset of the Hausdorff space L^L, hence closed; so $E(L,\gamma)$ consists of the iterates and ϕ. For each $P \in \mathcal{P}$, the map $P \circ \phi$ is the pointwise limit of the continuous functions $P \circ \gamma^n$, so it is of Baire class 1, and, by Proposition 5.1, the CCS is NIP. The space $E(L,\gamma)$ is a countable compact Hausdorff space, hence metrizable, so the CCS is NIP$_3$. Its members other than ϕ are continuous, which gives the statement about stability. □

5.2. **Stable and metrizable.** At the top of the hierarchy, the deep computations form a metrizable space. By Proposition 5.2, every convergent method lands here, and what separates stable from merely NIP$_3$ is whether the limit is continuous. A stable computation need not converge, however, as the first example shows.

A stable example: rotation. Let $L = \mathbb{T} = \mathbb{R}/\mathbb{Z}$, with the predicates $\cos 2\pi x$ and $\sin 2\pi x$, and let $\gamma(x) = x + \alpha$ with α irrational. A pointwise limit of rotations is a rotation, and the rotations $n\alpha$ are dense, so the deep computations are all the rotations $x \mapsto x + c$, $c \in \mathbb{T}$. They are continuous, so the rotation CCS is stable, and its space of deep computations is the circle group, a metrizable compactum.

NIP$_3$ but not stable: square roots. The iterates of the Newton map of Section 4.3 converge pointwise to $\mathfrak{f}$, which is discontinuous at 0 and ∞. By Proposition 5.2, the square-root CCS is NIP$_3$ but not stable.

NIP$_3$ but not stable: the power method. The power method computes a dominant eigenvector of a matrix by repeated multiplication and normalization. Let $A = \mathrm{diag}(\lambda_1,\dots,\lambda_d)$ with $\lambda_1 > \lambda_2 > \cdots > \lambda_d > 0$, let $L = \mathbb{RP}^{d-1}$, and let $\mathcal{P}$ consist of the predicates $p_{ij}([v]) = v_i v_j / \|v\|^2$, the entries of the orthogonal projection onto the line $[v]$. The step of the algorithm is $\gamma([v]) = [Av]$. For $[v] \in \mathbb{RP}^{d-1}$, let $i(v)$ be the least index with $v_i \neq 0$, and put $\pi([v]) = [e_{i(v)}]$.

Proposition 5.3. *The deep computations of the power method are $E(L,\gamma) = \{\gamma^n : n \in \mathbb{N}\} \cup \{\pi\}$. Hence the power method is NIP$_3$. Its only nontrivial deep computation π is of Baire class* 1 *and discontinuous, so the power method is not stable.*

Proof. For $i = i(v)$, we have $\lambda_i^{-n} A^n v = v_i e_i + \sum_{j>i} (\lambda_j/\lambda_i)^n v_j e_j \to v_i e_i$, so $\gamma^n \to \pi$ pointwise, and Proposition 5.2 applies. The map π is discontinuous at every $[v]$ with $i(v) > 1$, since $[v + te_1] \to [v]$ as $t \to 0$ while $\pi([v + te_1]) = [e_1] \neq \pi([v])$. □

The deep computation π, which returns *the dominant eigenvector of the input*, is not computable in the sense of Section 2, since it does not extend continuously to the space of types; yet it is the limit of a sequence of computations, and the closure in which it lives is as simple as possible.

Both levels: the logistic map. The logistic map $f_r(x) = rx(1-x)$, with $0 < r \leq 4$, maps $[0, 1]$ into itself. Its CCS has states $L = [0, 1]$, the single predicate $x \mapsto x$, and the semigroup of iterates of f_r. For $0 < r \leq 1$, let ℓ_r be the constant map 0; for $1 < r \leq 3$, let $\ell_r(x) = 1 - 1/r$ for $0 < x < 1$, and $\ell_r(0) = \ell_r(1) = 0$.

Proposition 5.4. *For $0 < r \leq 3$, the iterates of f_r converge pointwise to ℓ_r. Consequently the logistic CCS is stable for $0 < r \leq 1$, and NIP$_3$ but not stable for $1 < r \leq 3$.*

Proof. Write $f = f_r$ and $p = 1 - 1/r$, and note that $f(x)/x = r(1-x)$ for $x > 0$. If $r \leq 1$, then $f(x) < x$ for $0 < x \leq 1$, so every orbit decreases to a fixed point of f, and 0 is the only fixed point in $[0, 1]$.

Suppose that $1 < r \leq 2$. Then $0 < p \leq \frac{1}{2}$, and $f(x) > x$ for $0 < x < p$, while $f(x) < x$ for $p < x \leq 1$. For $0 < x < 1$ we have $f(x) \in (0, \frac{r}{4}] \subseteq (0, \frac{1}{2}]$, and f is increasing on $[0, \frac{1}{2}]$ and fixes p. Hence the orbit of $f(x)$ is increasing and bounded by p if $f(x) < p$, and decreasing and bounded below by p if $f(x) > p$. In both cases, and trivially if $f(x) = p$, it converges to a fixed point of f in $(0, \frac{1}{2}]$, that is, to p.

Suppose that $2 < r \leq 3$, and put $c = \frac{r}{4}$, the maximum of f, and $b = f(c)$. We have $p \leq c$, since $c - p = (r-2)^2/(4r)$, and therefore $b \leq f(p) = p$, since f is decreasing on $[\frac{1}{2}, 1]$. We also have $b \geq 1/r$, because $b - 1/r = (r^3(4-r) - 16)/(16r)$ and $r^3(4-r)$ increases from 16 on $[2, 3]$. The concave function f takes the value p at $1/r$ and at p, so $f \geq p$ on $[1/r, p]$. Hence f maps $J_- = [b, p]$ into $J_+ = [p, c]$, and, being decreasing on J_+, it maps J_+ onto J_-. Let $0 < x < 1$. The orbit of $f(x)$ lies in $(0, c]$, and it meets $J_- \cup J_+ = [b, c]$. Otherwise all its points lie in $(0, b)$, where $f(y) > y$ because $b < p$; they would then increase to a fixed point of f in $(0, b]$, but the only positive fixed point is p, and $p > b$ because f is strictly decreasing on $[\frac{1}{2}, 1]$ and $c > p$. Since f maps $J_- \cup J_+$ into itself, it remains to show that the orbit of every $y \in J_- \cup J_+$ converges to p. The points of period 2 of f that are not fixed are the roots of $r^2y^2 - r(r+1)y + (r+1)$, whose discriminant $r^2(r+1)(r-3)$ is negative for $r < 3$; for $r = 3$ the only root is p. Hence the function $f(f(y)) - y$ vanishes on $[0, 1]$ only at 0 and p, and it is negative on $(p, 1]$, since it is negative at 1. For $y \in J_+$ with $y > p$, the point $f(f(y))$ lies in J_+, so $p \leq f(f(y)) < y$. The sequence $(f^{2n}(y))$ therefore decreases to a fixed point of $f \circ f$ in $[p, y]$, that is, to p, and $f^{2n+1}(y) \to f(p) = p$. Every $y \in J_-$ satisfies $f(y) \in J_+$, so its orbit converges to p as well.

Finally, $f(0) = f(1) = 0$. In all cases the iterates converge pointwise to ℓ_r. The map ℓ_r is continuous for $r \leq 1$, and discontinuous at 0 for $r > 1$, so Proposition 5.2 gives the remaining statements. □

We return to the logistic family at $r = 4$, and to maps of the interval in general, at the end of the section.

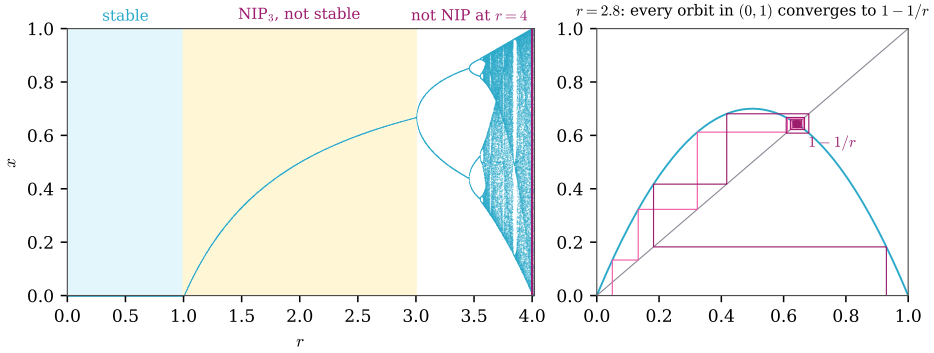


FIGURE 4. The logistic family $f_r(x) = rx(1-x)$. Left: the long-run values of the orbit of 0.3 for each r; the CCS is stable for $r \leq 1$, NIP$_3$ but not stable for $1 < r \leq 3$ (Proposition 5.4), and not NIP at $r = 4$ (Proposition 5.11); the parameters $3 < r < 4$ are not classified here. Right: for $r = 2.8$, two orbits converge to the fixed point $1 - 1/r$; the deep computation sends 0 and 1 to 0 and every other state to $1 - 1/r$.

5.3. **Hereditarily separable but not metrizable.** Below the metrizable level, the deep computations no longer settle down to a few limits: they form a family that a countable set of tests cannot separate (Proposition 3.4). Rounding produces such families, because an algorithm that rounds at every step makes a fresh decision at every step.

Drawing a line. Let $0 < \alpha < 1$ be irrational. To draw the line $y = \alpha x + \rho$ on a screen, Bresenham's algorithm [Bre65] moves one pixel to the right at each step and decides whether to move one pixel up. The n-th decision is $b_n(\rho) = \lfloor (n+1)\alpha + \rho \rfloor - \lfloor n\alpha + \rho \rfloor = [\![\{n\alpha + \rho\} \in [1-\alpha, 1)]\!]$, where $\{\cdot\}$ denotes the fractional part, and the state of the algorithm is the error term ρ modulo 1. The step $\rho \mapsto \rho + \alpha$ is a rotation of the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, but the decision bit is not a continuous function of ρ. The standard remedy [MH40, Lot02] is to take as states the decision sequences themselves.

Let $D = \{-k\alpha : k \in \mathbb{Z}\} \subseteq \mathbb{T}$, a countable dense set containing both endpoints 0 and $1 - \alpha$ of the decision interval. The *split circle* $\mathbb{T}_D$ is obtained from $\mathbb{T}$ by replacing each $t \in D$ by two points $t^- < t^+$, with the circular order topology; it is a compact metrizable space, homeomorphic to the Cantor set. The rotation by α preserves D and lifts to a homeomorphism R of $\mathbb{T}_D$ that preserves the labels $\pm$. The decision interval becomes clopen in $\mathbb{T}_D$, and the coding map $\mathbb{T}_D \to 2^{\mathbb{Z}}$, $y \mapsto ([\![R^n y \in [1-\alpha, 1)]\!])_{n \in \mathbb{Z}}$, is a homeomorphism onto the *Sturmian shift* X_α that conjugates R to the shift σ. The *Bresenham CCS* has states $L = X_\alpha$, the coordinate predicates, and the semigroup $\Gamma = \{\sigma^n : n \in \mathbb{N}\}$.

For $c \in \mathbb{T}$, let R_c^+ (respectively R_c^-) be the map of $\mathbb{T}_D$ that sends a point over y_0 to $(y_0 + c)^+$ (respectively $(y_0 + c)^-$) when $y_0 + c \in D$, and to $y_0 + c$ otherwise. The *double arrow circle* $\mathbb{D}$ is $\mathbb{T} \times \{-, +\}$ with the lexicographic circular order and its order topology; it is compact, first countable, hereditarily separable and not metrizable [Eng89, 3.10.C].

Theorem 5.5. *The Bresenham CCS is NIP. Its deep computations are*

$$E(X_\alpha, \sigma) = \{\sigma^n : n \in \mathbb{N}\} \cup \{R_c^+, R_c^- : c \in \mathbb{T}\},$$

where each σ^n is isolated and $c^\pm \mapsto R_c^\pm$ is a homeomorphism of $\mathbb{D}$ onto the remaining part. In particular, the Bresenham CCS is NIP$_2$ but not NIP$_3$.

Proof. For NIP, note that the predicates $P_j \circ \sigma^n$ are $\{0,1\}$-valued, and that the j-th coordinate of $\sigma^n y$ is 1 exactly when $R^{n+j}y$ lies in the clopen arc $[1-\alpha, 1)$ of $\mathbb{T}_D$, that is, when y lies in an arc of $\mathbb{T}_D$. A family of arcs of a circularly ordered set has Vapnik–Chervonenkis dimension at most 3, so it has no infinite independent subfamily.

For the deep computations, let γ be the limit of σ^{n_i} along an ultrafilter, and let $c = \lim n_i\alpha$ in $\mathbb{T}$. If n_i is eventually constant, then γ is a power of σ. Otherwise, since the points $n\alpha$ are distinct, the ultrafilter decides whether $n_i\alpha$ approaches c from the right or from the left. In the first case, for every point y over y_0 the points $R^{n_i}y$ over $y_0 + n_i\alpha$ approach $y_0 + c$ strictly from the right, and they converge in $\mathbb{T}_D$ to $(y_0 + c)^+$ if $y_0 + c \in D$ and to $y_0 + c$ otherwise; so $\gamma = R_c^+$. The second case gives R_c^-. Conversely, every c is approached from both sides by the points $n\alpha$ with $n \geq N$, for every N, so every $R_c^\pm$ is a deep computation, and $R_c^+ \neq R_c^-$, since they differ at the points over $d - c$ with $d \in D$. The same argument shows that $c_i^{\varepsilon_i} \to c^+$ in $\mathbb{D}$ implies $R_{c_i}^{\varepsilon_i} \to R_c^+$ pointwise, and symmetrically for c^-; so $c^\pm \mapsto R_c^\pm$ is a continuous bijection from the compact space $\mathbb{D}$ onto a Hausdorff space, hence a homeomorphism. Finally, fix $d \in D$ and consider the two points $(d - n\alpha)^\pm$ of $\mathbb{T}_D$, which σ^n sends to d^- and d^+. Every map $R_c^\pm$ sends them to the same point, and so does every pointwise limit of σ^{m_i} with $m_i \to \infty$, by the argument above. Hence no net of such maps, and no net of maps σ^m with $m \neq n$ and m bounded, converges to σ^n, so σ^n is isolated. Thus $E(X_\alpha, \sigma)$ is the union of a countable set and a copy of $\mathbb{D}$; it is hereditarily separable and not metrizable. □

The two deep computations R_c^+ and R_c^- are the two ways of drawing the line infinitely far away, where the accumulated error has reached c: they agree except at the pixels where the line passes exactly through a corner, and there they apply the two opposite rounding conventions. Figure 5 shows the two conventions.

The space $E(X_\alpha, \sigma)$ is the closure of a countable set, so it is a separable Rosenthal compactum; it is hereditarily separable and not metrizable, so, by Todorčević's trichotomy (Theorem 3.3), it contains a copy of the split Cantor set $S(2^{\mathbb{N}})$. For the action of $\mathbb{Z}$ generated by the shift, Glasner and Megrelishvili proved that Sturmian systems are tame$_2$ but not hereditarily non-sensitive [GM22, Corollary 8.3].

5.4. **First countable but not hereditarily separable.** Running two rounding algorithms side by side moves the computation one level down.

Drawing a line in space. Let α and β be irrational numbers in $(0,1)$ such that $1, \alpha, \beta$ are linearly independent over $\mathbb{Q}$. To draw a line in $\mathbb{Z}^3$ with direction $(1, \alpha, \beta)$, the three-dimensional Bresenham algorithm runs the planar algorithm in two coordinates at once. Its CCS has states $L = X_\alpha \times X_\beta$, the coordinate predicates of both factors, and the semigroup generated by $\sigma \times \sigma$.

Theorem 5.6. *The three-dimensional Bresenham CCS is NIP$_1$ but not NIP$_2$.*

Proof. Each predicate is a coordinate of one factor, so the CCS is NIP by Theorem 5.5. The deep computations form a closed subspace of $E(X_\alpha, \sigma) \times E(X_\beta, \sigma)$,

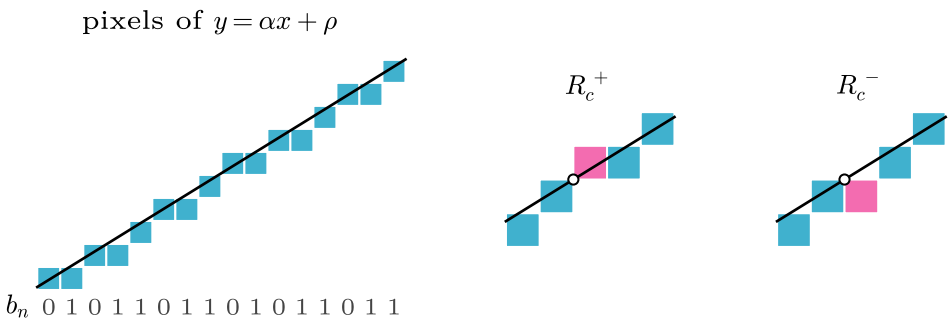


FIGURE 5. Left: the pixels drawn by Bresenham's algorithm for $\alpha = (\sqrt{5}-1)/2$, with the decision bits b_n. Right: where the line passes exactly through a pixel corner, the deep computations R_c^+ and R_c^- break the tie in opposite directions; they agree on every other pixel.

which is first countable as a product of two first countable spaces; so the CCS is NIP_1. By Kronecker's theorem, the points $(n\alpha, n\beta)$ with $n \geq N$ are dense in $\mathbb{T}^2$ for every N, so every point (c, c') is approached by them from inside each of the four open quadrants at (c, c'); arguing as in the proof of Theorem 5.5, every pair $(R_c^{\varepsilon}, R_{c'}^{\varepsilon'})$ is a deep computation. Consider the pairs (R_c^+, R_{-c}^+), for $c \in \mathbb{T}$. In $\mathbb{D} \times \mathbb{D}$, a neighborhood of $(c^+, (-c)^+)$ is $[c^+, (c+\delta)^-) \times [(-c)^+, (-c+\delta)^-)$, and it contains $(s^+, (-s)^+)$ only if $c \leq s < c + \delta$ and $-c \leq -s < -c + \delta$, that is, only if $s = c$. So these pairs form an uncountable discrete subspace, and the space of deep computations is not hereditarily separable. □

The uncountable discrete set consists of the deep computations that break ties upward in both coordinates at points where the two accumulated errors are opposite. Each of the two planar algorithms is NIP_2, and drawing in two directions at once leaves the class: like the Sorgenfrey plane, a product of two hereditarily separable spaces need not be hereditarily separable. Compare [GM22, Proposition 4.1], where a product of two tame_2 systems is shown not to be tame_2. By Todorčević's trichotomy, the space of deep computations contains a copy of $D(2^{\mathbb{N}})$ or of $\hat{D}(S(2^{\mathbb{N}}))$.

5.5. **NIP but not first countable: the Euclidean algorithm.** At the last tame level, some deep computation is not determined by countably many tests of its own (Proposition 3.4). An exact equality test has this property: a countable set of inputs cannot tell it apart from the test that always fails. The Euclidean algorithm produces such tests.

Let $L = \mathbb{RP}^1 = \mathbb{R} \cup \{\infty\}$, with the predicates P_1, P_2 of Section 4.3, and let $\Gamma = \mathrm{PGL}(2, \mathbb{Z})$ be the group of Möbius maps $x \mapsto (ax + b)/(cx + d)$ with integer coefficients and $ad - bc = \pm 1$. It is generated by the steps $x \mapsto x + 1$ and $x \mapsto 1/x$ of the Euclidean algorithm, and the maps that the continued fraction algorithm composes are elements of Γ. We call $(\mathbb{RP}^1, (P_1, P_2), \Gamma)$ the *Euclidean CCS*. It is a dynamical CCS whose semigroup is a group, and its deep computations form $E(\mathbb{RP}^1, \Gamma)$.

For an irrational number $\eta = [a_0; a_1, a_2, \dots]$, let p_k/q_k be its convergents, put $p_{-1} = 1$ and $q_{-1} = 0$, and let $\eta_{k+1} = [a_{k+1}; a_{k+2}, \dots]$ be its complete quotients. Let M_k be the Möbius map $t \mapsto (p_k t + p_{k-1})/(q_k t + q_{k-1})$, which belongs to Γ because $p_k q_{k-1} - p_{k-1} q_k = (-1)^{k+1}$. Writing $A_j(t) = a_j + 1/t$, we have $M_k = A_0 \circ A_1 \circ \cdots \circ A_k$, so $M_k^{-1} = A_k^{-1} \circ \cdots \circ A_0^{-1}$ with $A_j^{-1}(t) = 1/(t - a_j)$. Thus M_k^{-1} is the map that the first $k+1$ steps of the continued fraction algorithm for η apply, and the classical identity $\eta = M_k(\eta_{k+1})$ says that it sends η to its complete quotient η_{k+1}.

Lemma 5.7. *Let η be irrational, and let $x \in \mathbb{RP}^1$ with $x \neq \eta$. Then $M_k^{-1}(x) + q_{k-1}/q_k \to 0$ as $k \to \infty$, while $M_k^{-1}(\eta) = \eta_{k+1} > 1$.*

Proof. Solving $M_k(t) = x$ gives $M_k^{-1}(\infty) = -q_{k-1}/q_k$ and, for real x,

$$M_k^{-1}(x) = \frac{q_{k-1}x - p_{k-1}}{p_k - q_k x} = -\frac{q_{k-1}}{q_k} \cdot \frac{x - p_{k-1}/q_{k-1}}{x - p_k/q_k}$$

for $k \geq 1$ (so that $q_{k-1} \geq 1$), where the value is ∞ if $x = p_k/q_k$. The convergents of an irrational number are pairwise distinct, so x equals p_k/q_k for at most one k; for all larger k the formula gives a real number. Both convergents tend to $\eta \neq x$, so the last fraction tends to 1, and $0 < q_{k-1}/q_k \leq 1$ for $k \geq 1$; hence $M_k^{-1}(x) + q_{k-1}/q_k \to 0$. The identity $M_k^{-1}(\eta) = \eta_{k+1}$ is the classical one, and $\eta_{k+1} > a_{k+1} \geq 1$ because η_{k+1} is irrational. □

Let $\varphi = (1 + \sqrt{5})/2$. Let S be the set of irrational numbers $\eta = [0; b_1, b_2, \dots]$ whose sequence of partial quotients is a concatenation $1^1 d_1 1^2 d_2 1^3 d_3 \cdots$, where 1^n denotes n consecutive ones and each $d_n \in \{1, 2\}$. The block 1^n occupies the indices $s_n, \dots, s_n + n - 1$ with $s_n = n(n+1)/2$. Distinct choices of (d_n) give distinct numbers, by the uniqueness of continued fraction expansions, so S has the cardinality of the continuum.

Proposition 5.8. *(i) For every $\eta \in S$, the map δ_η that sends η to φ and every other point of $\mathbb{RP}^1$ to $-1/\varphi$ is a deep computation of the Euclidean CCS.*

(ii) The constant map c with value $-1/\varphi$ is a deep computation, and it has no countable base of neighborhoods in $E(\mathbb{RP}^1, \Gamma)$.

Consequently $E(\mathbb{RP}^1, \Gamma)$ is not first countable.

Proof. (i) Let $\eta \in S$, and for each n let $k_n = s_n + \lfloor n/2 \rfloor$ be the middle index of the block 1^n, and $m_n = \lceil n/2 \rceil - 1$. Then $a_{k_n} = a_{k_n - 1} = \cdots = a_{k_n - m_n} = 1$ and $a_{k_n + 1} = \cdots = a_{k_n + m_n} = 1$, and $m_n \to \infty$. We have $q_{k-1}/q_k = [0; a_k, a_{k-1}, \dots, a_1]$ and $\eta_{k+1} = [a_{k+1}; a_{k+2}, \dots]$. Suppose that two continued fractions, finite or infinite, $[c_0; c_1, \dots]$ and $[c_0'; c_1', \dots]$ satisfy $c_j = c_j'$ for $j < m$, with $c_1 = \cdots = c_{m-1} = 1$. Then they share the convergents of index at most $m - 1$, and both lie between the last two of them (or equal one of them). So they differ by at most $1/(q_{m-2} q_{m-1}) \leq 1/(F_{m-1} F_m)$, where F_j are the Fibonacci numbers, whatever the remaining partial quotients are. Both $[0; a_{k_n}, \dots, a_1]$ versus $[0; 1, 1, \dots]$ and $[a_{k_n+1}; a_{k_n+2}, \dots]$ versus $[1; 1, 1, \dots]$ satisfy this with $m = m_n$. Hence $q_{k_n - 1}/q_{k_n} \to [0; 1, 1, \dots] = 1/\varphi$ and $\eta_{k_n + 1} \to [1; 1, 1, \dots] = \varphi$. By Lemma 5.7, the maps $M_{k_n}^{-1} \in \Gamma$ converge pointwise to δ_η, at $x = \infty$ as well.

(ii) Let (η_j) be a sequence of distinct elements of S. For each $x \in \mathbb{RP}^1$, at most one j has $\eta_j = x$, so $\delta_{\eta_j}(x) = -1/\varphi$ for all but at most one j. Hence $\delta_{\eta_j} \to c$ pointwise, and $c \in E(\mathbb{RP}^1, \Gamma)$ because this set is closed. Now let $(U_i)_{i \in \mathbb{N}}$

be open neighborhoods of c in $E(\mathbb{RP}^1, \Gamma)$. Each U_i contains a basic neighborhood $\{f : d(f(x), c(x)) < \varepsilon_i \text{ for } x \in F_i\}$ with $F_i \subseteq \mathbb{RP}^1$ finite, $\varepsilon_i > 0$ and d a metric on $\mathbb{RP}^1$. The set $F = \bigcup_i F_i$ is countable, so some $\eta \in S$ lies outside F, and then δ_η agrees with c on F. Hence $\delta_\eta \in \bigcap_i U_i$, while $\delta_\eta \neq c$. Thus no countable family of neighborhoods of c has intersection $\{c\}$. In a Hausdorff space, the intersection of a base of neighborhoods of c is $\{c\}$, so c has no countable base of neighborhoods. □

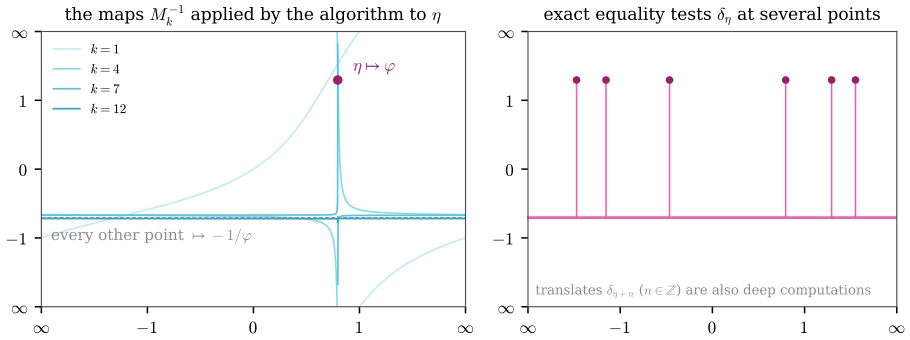


FIGURE 6. The Euclidean algorithm, drawn on the projective line (each axis runs from ∞ through 0 back to ∞). Left: the maps M_k^{-1} that the continued fraction algorithm applies to a number η whose partial quotients contain longer and longer runs of ones; as k grows along the middles of the runs, every point other than η is sent near $-1/\varphi$, while η is sent near φ. Right: the limits, exact equality tests δ_η, at several points. Every neighborhood of the constant map $-1/\varphi$ contains all but finitely many of the uncountably many tests δ_η, so the space of deep computations is not first countable (Proposition 5.8).

Corollary 5.9. *The Euclidean CCS is NIP but not NIP$_1$. Its space of deep computations contains the copy $\{\delta_\eta : \eta \in S\} \cup \{c\}$ of the Alexandroff compactification $A(2^{\mathbb{N}})$.*

Proof. Ellis proved that the action of $\mathrm{GL}(2, \mathbb{R})$ on the projective line is tame [Ell93]; see [GM22, Example 5.7]. For every continuous $f\colon \mathbb{RP}^1 \to \mathbb{R}$, the family $\{f \circ \gamma : \gamma \in \Gamma\}$ is contained in the corresponding family for $\mathrm{GL}(2, \mathbb{R})$, so it has the NIP. Hence the action of Γ is tame, and, by Proposition 5.1, the Euclidean CCS is NIP. By Proposition 5.8, it is not NIP$_1$. Each δ_η is isolated in $\{\delta_\eta : \eta \in S\} \cup \{c\}$, since it is the only member that sends η near φ, and every neighborhood of c contains all but finitely many of the δ_η, as in the proof of Proposition 5.8. This is the Alexandroff compactification of a discrete set of cardinality $2^{\aleph_0}$, which is homeomorphic to $A(2^{\mathbb{N}})$ (Section 1.4). □

The deep computation δ_η is an exact equality test: it answers one value if the input is exactly η and another value otherwise. It is the limit of the maps that the continued fraction algorithm applies to η, evaluated at all other inputs. The mechanism is that of convergence groups. By Lemma 5.7, the maps M_k^{-1} draw

the images of all points other than η together, near $-q_{k-1}/q_k \in [-1, 0)$, while they send η into $(1, \infty)$; this exhibits every irrational η as a conical limit point of Γ. The choice of S makes the two limits the same for uncountably many η. Glasner and Megrelishvili use the same device to prove that an infinite countable uniform convergence group acting on a compact metric space with infinite limit set is tame but not tame_1 [GM22, Proposition 6.1 and Theorem 6.5]. Their theorem does not cover Γ, which contains parabolic elements and is not a uniform convergence group on $\mathbb{RP}^1$. Ellis's example, the projective action of the whole group $\mathrm{GL}(2, \mathbb{R})$, is also tame but not tame_1 [GM22, Example 5.7]; the Euclidean CCS shows that the countable group generated by the two steps of the algorithm already reaches this level.

A single step cannot produce δ_η. For $g \in \Gamma$, the closure of $\{g^n : n \in \mathbb{Z}\}$ in $(\mathbb{RP}^1)^{\mathbb{RP}^1}$ is countable, hence metrizable, and it does not contain δ_η. Indeed, if g has finite order, the set of its powers is finite. If g is parabolic, with fixed point r, then g^n converges pointwise to the constant map r as $n \to \pm\infty$. If g is hyperbolic, with attracting and repelling fixed points a and r, then g^n converges to the map that fixes r and sends every other point to a as $n \to +\infty$, and to the map that fixes a and sends every other point to r as $n \to -\infty$. If g reverses orientation and has infinite order, then its trace t is nonzero, so g^2 has trace $t^2+2 > 2$ and is hyperbolic, with the same fixed points as g; the odd powers $g^{2m+1} = g \circ g^{2m}$ have the same two limits as the even ones. In every case the closure consists of the powers of g, which are homeomorphisms, and at most two limit maps, each of which is constant or fixes the unique point at which it differs from a constant. The map δ_η is none of these, since it is not injective and $\delta_\eta(\eta) = \varphi \neq \eta$.

Free reduction. A second example at this level is the free group F_2 on two generators a, b, acting on its boundary ∂F_2, the space of infinite reduced words in a, a^{-1}, b, b^{-1}, which is a Cantor set. A generator acts on an infinite word by prepending a letter and cancelling it against the first letter if they are inverse, as a stack does. With the predicates $w \mapsto [\![w_n = x]\!]$, for $n \in \mathbb{N}$ and each letter x, this is a dynamical CCS, and Glasner and Megrelishvili proved that the system is tame but not tame_1 [GM22, Corollary 6.6 and Example 6.7]. By Proposition 5.1, this CCS is NIP but not NIP_1.

An open question. Both examples at this level are actions of groups with two generators. Whether a single map can reach this level, that is, whether some cascade CCS is NIP but not NIP_1, is Question 7.2. For actions of $\mathbb{Z}$, and more generally of abelian groups, the existence of a tame metrizable system whose enveloping semigroup is not first countable is Question 6.8 of Glasner and Megrelishvili [GM22]; to our knowledge, it is open.

5.6. **Beyond NIP.** Beyond the hierarchy lie the computations with the independence property: for some feature, the computations realize every finite pattern of high and low values on suitable inputs. Chaotic iteration produces such patterns.

Newton's method on the Riemann sphere. On its Julia set the Newton map is chaotic, and the chaos is visible to the predicates of the CCS. We exhibit the independence in the simplest case, the square roots of 1.

Proposition 5.10. *Let $p(z) = z^2 - 1$, and let $N(z) = \frac{1}{2}\big(z + \frac{1}{z}\big)$ be its Newton map on $\hat{\mathbb{C}}$. The functions $P_2 \circ N^{3k}$, $k \in \mathbb{N}$, form an independent sequence: for all*

finite disjoint sets $E, F \subseteq \mathbb{N}$ there is $z \in \hat{\mathbb{C}}$ with $P_2(N^{3k}z) \geq \frac{1}{2}$ for $k \in E$ and $P_2(N^{3k}z) \leq -\frac{1}{2}$ for $k \in F$. In particular, the Newton CCS of p is not NIP.

Proof. The Möbius transformation $\phi(z) = (z-1)/(z+1)$ conjugates N to the squaring map: $\phi(N(z)) = \phi(z)^2$. It maps the imaginary axis, together with ∞, onto the unit circle, with $\phi^{-1}(e^{i\theta}) = i\cot(\theta/2)$. Hence, for $z = i\cot(\theta/2)$, we have $N^n(z) = i\cot(2^n\theta/2)$ and

$$P_2\big(N^n(z)\big) = \frac{2\cot(2^{n-1}\theta)}{\cot^2(2^{n-1}\theta)+1} = \sin(2^n\theta).$$

Write $\theta = 2\pi\sum_{j\geq 1} x_j 2^{-j}$ with bits x_j. The fractional part of $2^{3k}\theta/2\pi$ has binary expansion $0.x_{3k+1}x_{3k+2}x_{3k+3}\cdots$. If $(x_{3k+1}, x_{3k+2}, x_{3k+3}) = (0,1,0)$, this fractional part lies in $[\frac{1}{4}, \frac{3}{8}]$, so $\sin(2^{3k}\theta) \geq \sin\frac{3\pi}{4} > \frac{1}{2}$; if the block is $(1,1,0)$, it lies in $[\frac{3}{4}, \frac{7}{8}]$, so $\sin(2^{3k}\theta) < -\frac{1}{2}$. Choosing the block $(0,1,0)$ for $k \in E$ and $(1,1,0)$ for $k \in F$ gives the required z. □

The proof also shows that the doubling map $x \mapsto 2x$ of the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, with the predicates $\cos 2\pi x$ and $\sin 2\pi x$, is not NIP: with $x = \theta/2\pi$, the functions $\sin(2\pi\, 2^{3k}x)$ form an independent sequence. The doubling map shifts the binary digits of x, and the independence comes from the freedom to choose those digits.

The same algorithm is tame on the real line. The Julia set of N is the imaginary axis together with ∞, which meets $\mathbb{RP}^1$ only at ∞, and on $\mathbb{RP}^1$ the iterates converge to the deep computation of Section 4.3. Whether Newton's method is tame depends on the space of states, not on the formula for the step.

The logistic map at $r = 4$. The doubling map also governs the logistic map $f_4(x) = 4x(1-x)$ of Section 5.2.

Proposition 5.11. *The functions f_4^{3k}, $k \in \mathbb{N}$, form an independent sequence on $[0,1]$: for all finite disjoint sets $E, F \subseteq \mathbb{N}$ there is $x \in [0,1]$ with $f_4^{3k}(x) \geq \frac{3}{4}$ for $k \in E$ and $f_4^{3k}(x) \leq \frac{1}{4}$ for $k \in F$. In particular, the logistic CCS for $r = 4$ is not NIP.*

Proof. Let $h(\theta) = \sin^2(\pi\theta)$ for $\theta \in \mathbb{T}$. Then $f_4(h(\theta)) = 4\sin^2(\pi\theta)\cos^2(\pi\theta) = h(2\theta)$, so $f_4^n(h(\theta)) = \sin^2(\pi\, 2^n\theta)$. Write $\theta = \sum_{j\geq 1} x_j 2^{-j}$ in binary. The fractional part u of $2^{3k}\theta$ is then $0.x_{3k+1}x_{3k+2}\cdots$ in binary. If $(x_{3k+1}, x_{3k+2}, x_{3k+3}) = (0,1,1)$, then $u \in [\frac{3}{8}, \frac{1}{2}]$ and $\sin^2(\pi u) \geq \sin^2\frac{3\pi}{8} > \frac{3}{4}$; if the block is $(0,0,0)$, then $u \in [0, \frac{1}{8}]$ and $\sin^2(\pi u) \leq \sin^2\frac{\pi}{8} < \frac{1}{4}$. Choosing the block $(0,1,1)$ for $k \in E$ and $(0,0,0)$ for $k \in F$, and putting $x = h(\theta)$, gives the required point. □

Thus the logistic family, which is stable for $0 < r \leq 1$ and NIP_3 for $1 < r \leq 3$ (Proposition 5.4), leaves the hierarchy at $r = 4$. On the interval, the tame levels between NIP_3 and NIP are empty. A continuous map of a compact interval is tame if and only if it is not chaotic in the sense of Li and Yorke [Li11, Theorems 4.23 and 4.24], and Glasner and Megrelishvili record, from an unpublished paper, that it is tame if and only if it is hereditarily non-sensitive [GM23, Remark 5.4]. Thus, for cascade CCSs on $[0,1]$ with the predicate $x \mapsto x$, the levels NIP, NIP_1, NIP_2 and NIP_3 coincide.

The Thue–Morse sequence. Even a sequence produced by a simple substitution rule can be wild. The Thue–Morse sequence t_n is the parity of the number of ones in the binary expansion of n; it is the limit of the words obtained from 0 by the substitution $0 \mapsto 01$, $1 \mapsto 10$. The two-sided sequences all of whose finite blocks

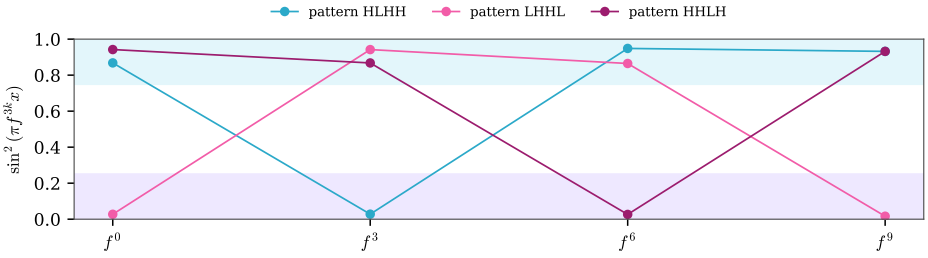


FIGURE 7. Independence for the doubling map. The values of the predicate $\sin^2(\pi x)$ at the times $0, 3, 6, 9$ follow any prescribed pattern of high (H, at least $3/4$) and low (L, at most $1/4$): the initial state is chosen with the block 011 of binary digits for H and 000 for L.

occur in (t_n) form the Thue–Morse subshift $X_{TM} \subseteq 2^{\mathbb{Z}}$, on which the shift σ is a homeomorphism. Glasner and Megrelishvili note that the action of $\mathbb{Z}$ that it generates is not tame [GM23, Example 5.3]. So the CCS with states X_{TM}, the coordinate predicates, and the group $\{\sigma^n : n \in \mathbb{Z}\}$ is not NIP, by Proposition 5.1. The cascade CCS of σ is not NIP either. Indeed, the words obtained from 0 by an even number of applications of the substitution are palindromes, and every finite block of (t_n) occurs in one of them, so the reversal ρ of two-sided sequences maps X_{TM} onto itself and satisfies $\rho \circ \sigma = \sigma^{-1} \circ \rho$. An independent sequence $f \circ \sigma^{n_k}$, with f continuous and the n_k distinct, has infinitely many indices n_k of the same sign. If they are positive, these terms already form an independent sequence in the cascade; if they are negative, the functions $f \circ \sigma^{n_k} \circ \rho = (f \circ \rho) \circ \sigma^{-n_k}$ form an independent sequence with positive exponents, because ρ is a bijection of X_{TM}. By Proposition 5.1, the cascade CCS is not NIP.

5.7. **Summary.** Table 2 collects the examples of this section. The Euclidean algorithm and free reduction are actions of groups with two generators; every other entry is a single map iterated.

6. Randomized versions of NIP and Monte Carlo computability of deep computations

Sections 3–5 studied deep computations as *topological* limits: pointwise limits of computations, of the first Baire class on the type shards, and approximated by sequences of computations. That theory rests on the type shards being Polish, and hence on the assumption that the set $\mathcal{P}$ of predicates is countable (it is used in the proof of Theorem 1.9). In this section we drop that assumption and replace topology by measure: instead of asking that a deep computation be of the first Baire class, we ask that it be *universally measurable*, that is, measurable with respect to every Radon probability measure on the type shards, so that it can be evaluated almost surely by random sampling. We call such deep computations Monte Carlo computable. The role of the Bourgain–Fremlin–Talagrand dichotomy is taken by its measure-theoretic form (Theorem 6.5), and that of NIP is shared with Talagrand stability.

Level	Computation	Deep computations; source
stable	rotation of the circle	the circle group
	logistic map, $0 < r \leq 1$	iterates and one continuous limit; Prop. 5.4
NIP_3, not stable	power method	countable; Prop. 5.3
	Newton's method for $x^2 - a$ on $\mathbb{RP}^1$	countable; Prop. 5.2
	logistic map, $1 < r \leq 3$	countable; Prop. 5.4
NIP_2, not NIP_3	Bresenham, plane (Sturmian shift)	isolated iterates and a double arrow circle; Thm. 5.5
NIP_1, not NIP_2	Bresenham, space (two Sturmian shifts)	contains an uncountable discrete set; Thm. 5.6
NIP, not NIP_1	Euclidean algorithm: $\mathrm{PGL}(2,\mathbb{Z})$ on $\mathbb{RP}^1$ (group)	contains $A(2^{\mathbb{N}})$, of exact equality tests; Cor. 5.9
	free reduction: F_2 on ∂F_2 (group)	not first countable; [GM22]
	a single map	open; Question 7.2
not NIP	doubling map of the circle	Prop. 5.10
	Newton's method for $z^2 - 1$ on $\hat{\mathbb{C}}$	Prop. 5.10
	logistic map, $r = 4$	Prop. 5.11
	Thue–Morse shift	[GM23]

TABLE 2. Computations at each level of the hierarchy.

The main results of the section are Theorem 6.7, which connects NIP and Monte Carlo computability, and Corollary 6.14, which connects Talagrand stability and Monte Carlo computability.

6.1. **NIP and Monte Carlo computability of deep computations.** By Fact 1.2, for perfectly normal X, a function f is in $\mathrm{B}_1(X,Y)$ if and only if $f^{-1}[U]$ is an F_σ subset of X for every open $U \subseteq Y$. This motivates the following definition.

Definition 6.1. Given a Hausdorff space X and a measurable space (Y,Σ), we say that $f : X \to Y$ is *universally measurable* (with respect to Σ) if $f^{-1}(E)$ is μ-measurable for every Radon measure μ on X and every $E \in \Sigma$. When $Y = \mathbb{R}$, we will always take $\Sigma = \mathcal{B}(\mathbb{R})$, the Borel σ-algebra of $\mathbb{R}$.

If X is a compact (Hausdorff) space, then Radon measures on X are finite, and hence, through renormalization, we can regard them as probability measures. We thus have the following remark:

Remark 6.2. If X is compact, then a function $f : X \to \mathbb{R}$ is universally measurable if and only if $f^{-1}(U)$ is μ-measurable for every Radon probability measure μ on X and every open set $U \subseteq \mathbb{R}$.

Intuitively, a function is universally measurable if it is "measurable no matter which reasonable way you try to measure things on its domain". The concept of

universal measurability emerged from work of Kallianpur and Sazonov, in the late 1950s and 1960s — with later developments by Blackwell, Darst and others — building on earlier ideas of Gnedenko and Kolmogorov from the 1950s. See [Pap02, Chapters 1 and 2].

Notation 6.3. Following [BFT78], the collection of all universally measurable real-valued functions on X will be denoted by $M_r(X)$. Given a fixed Radon measure μ on X, the collection of all μ-measurable real-valued functions on X will be denoted by $\mathscr{M}^0(X,\mu)$.

In the context of deep computations, we are interested in transition maps of a state space $L \subseteq \mathbb{R}^{\mathcal{P}}$ into itself. In the product space $\mathbb{R}^{\mathcal{P}}$, we have two natural σ-algebras: the Borel σ-algebra, i.e., the σ-algebra generated by open sets in $\mathbb{R}^{\mathcal{P}}$, and the cylinder σ-algebra (i.e., the σ-algebra generated by the sub-basic open sets in $\mathbb{R}^{\mathcal{P}}$, that is, the set $\pi_i^{-1}(U)$ with $U \subseteq \mathbb{R}$ open). In general, the cylinder σ-algebra is strictly smaller; however, when $\mathcal{P}$ is countable, both σ-algebras coincide. We will use the cylinder σ-algebra to define universally measurable maps $f : \mathbb{R}^{\mathcal{P}} \to \mathbb{R}^{\mathcal{P}}$. The reason for this choice is the following characterization:

Proposition 6.4. *Let X be a Hausdorff space and $Y = \prod_{i\in I} Y_i$ be any product of measurable spaces (Y_i, Σ_i) for $i \in I$. Let Σ_Y be the cylinder σ-algebra generated by the measurable spaces (Y_i, Σ_i). The following are equivalent for $f : X \to Y$:*

(i) $f : X \to Y$ is universally measurable (with respect to Σ_Y).
(ii) $\pi_i \circ f : X \to Y_i$ is universally measurable (with respect to Σ_i) for all $i \in I$.

Proof. (i)⇒(ii) is clear since the projection maps π_i are measurable and the composition of measurable functions is measurable. To prove (ii)⇒(i), note that if $C = \prod_{i\in I} C_i$ is a measurable cylinder and J is the finite subset of I such that $C_i \neq Y_i$ for $i \in J$, then, $f^{-1}(C) = \bigcap_{i\in J}(\pi_i \circ f)^{-1}(C_i)$ is universally measurable by assumption. □

The preceding proposition says that we can check measurability of a transition just by checking measurability feature by feature. This is analogous to the Baire class 1 setting (see Proposition 1.8).

We saw in section 1 that, when the set $\mathcal{P}$ of predicates is countable, PAC-learning (or NIP) corresponds to relative compactness in the space of Baire class 1 functions (see Theorems 1.4, 1.6, 1.9). The following result (which does not assume countability of $\mathcal{P}$) gives an analogous characterization of the NIP in terms of universal measurability:

Theorem 6.5 (Bourgain–Fremlin–Talagrand, Theorem 2F in [BFT78])**.** *Let X be a Hausdorff space and $A \subseteq \mathrm{C}(X)$ be pointwise bounded. The following are equivalent:*

(i) $\overline{A} \subseteq M_r(X)$.
(ii) For every compact $K \subseteq X$, $A|_K$ satisfies the NIP.
(iii) For every Radon measure μ on X, A is relatively countably compact in $\mathscr{M}^0(X,\mu)$, i.e., every countable subset of A has a limit point in $\mathscr{M}^0(X,\mu)$.

This result allows us to formalize the concept of a deep computation being *Monte Carlo computable*, which we define below (Definition 6.6). To motivate the definition, let us first recall two facts:

(1) Littlewood's second principle states that every Lebesgue measurable function is "nearly continuous". The formal statement of this, which is Luzin's

theorem, is that if (X, Σ, μ) a Radon measure space and Y is a second-countable topological space (e.g., $Y = \mathbb{R}^{\mathcal{P}}$ with $\mathcal{P}$ countable) equipped with the Borel σ-algebra, then any given $f : X \to Y$ is measurable if and only if for every $E \in \Sigma$ and every $\varepsilon > 0$ there exists a closed $F \subseteq E$ such that the restriction $f|_F$ is continuous and $\mu(E \backslash F) < \varepsilon$.

(2) Computability of deep computations is characterized in terms of continuous extendibility of computations. This is at the core of [ADIW26].

These two facts motivate the following definition:

Definition 6.6. Let $(L, \mathcal{P}, \Gamma)$ be a CCS. We say that a transition $f : L \to L$ is *universally Monte Carlo computable* if and only if there exists an extension $\tilde{f} : \mathcal{L}_{\mathrm{sh}} \to \mathcal{L}_{\mathrm{sh}}$ of f satisfying the following property: for every sizer $r_\bullet$ there is a sizer $s_\bullet$ such that the restriction $\tilde{f}|_{\mathcal{L}[r_\bullet]} : \mathcal{L}[r_\bullet] \to \mathcal{L}[s_\bullet]$ is universally measurable, i.e., $\pi_P \circ \tilde{f}|_{\mathcal{L}[r_\bullet]} : \mathcal{L}[r_\bullet] \to [-s_P, s_P]$ is μ-measurable for every Radon probability measure μ on $\mathcal{L}[r_\bullet]$ and $P \in \mathcal{P}$.

Note that, since each type-shard $\mathcal{L}[r_\bullet]$ is compact, by Remark 6.2, in the context of Monte Carlo computability, we restrict our attention to Radon probability measures instead of general Radon measures.

Theorem 6.7. *Let $(L, \mathcal{P}, \Gamma)$ be a CCS satisfying the Extendibility Axiom. Let R be an exhaustive collection of sizers. Let $\Delta \subseteq \Gamma$ be R-confined. If $\pi_P \circ \Delta|_{L[r_\bullet]}$ satisfies the NIP for all $P \in \mathcal{P}$ and all $r_\bullet \in R$, then every deep computation in $\overline{\Delta}$ is universally Monte Carlo computable.*

Proof. Fix $P \in \mathcal{P}$ and $r_\bullet \in R$. By the Extendibility Axiom, $\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}$ is a set of pointwise bounded continuous functions on the compact set $\mathcal{L}[r_\bullet]$. By Lemma 1.10, $\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}$ has the NIP, hence so does its restriction to every compact $K \subseteq \mathcal{L}[r_\bullet]$, and Theorem 6.5 gives $\overline{\pi_P \circ \tilde{\Delta}|_{\mathcal{L}[r_\bullet]}} \subseteq M_r(\mathcal{L}[r_\bullet])$. If $f \in \overline{\Delta}$ and $\tilde{f}$ is the extension of Lemma 2.7, then $\pi_P \circ \tilde{f}|_{\mathcal{L}[r_\bullet]}$ belongs to this closure. □

Whether the converse of Theorem 6.7 holds is Question 7.3.

6.2. **Talagrand stability and Monte Carlo computability of deep computations.** There is another notion closely related to NIP, introduced by Talagrand in [Tal84] while studying Pettis integration. Suppose that X is a compact Hausdorff space and $A \subseteq \mathbb{R}^X$. Let μ be a Radon probability measure on X. Given a μ-measurable set $E \subseteq X$, a positive integer k and real numbers $a < b$, let

$$D_k(A, E, a, b) = \bigcup_{f \in A} \{x \in E^{2k} : f(x_{2i}) \leq a,\ f(x_{2i+1}) \geq b \ \text{ for all } i < k\}.$$

Definition 6.8. We say that A is *Talagrand μ-stable* if and only if for every μ-measurable set $E \subseteq X$ of positive measure and for every $a < b$ there is $k \geq 1$ such that

$$(\mu^{2k})^*(D_k(A, E, a, b)) < (\mu(E))^{2k},$$

where μ^* denotes the outer measure.

Note that the reference to outer measure in Definition 6.8 is necessary since $D_k(A, E, a, b)$ need not be measurable. The inequality certainly holds when A is a countable set of continuous (or μ-measurable) functions.

Remarks 6.9. Let X, μ, $A \subseteq \mathbb{R}^X$, and $E \subseteq X$ be as above. Then,

(1) If $A \subseteq B \subseteq \mathbb{R}^X$, then
$$D_k(A, E, a, b) \subseteq D_k(B, E, a, b).$$
(2) If $a < a' < b' < b$, then
$$D_k(\overline{A}, E, a, b) \subseteq D_k(A, E, a', b').$$
[*Proof*: If $x \in D_k(\overline{A}, E, a, b)$, then there is $f \in \overline{A}$ such that $f(x_{2i}) \leq a < a'$ and $f(x_{2i+1}) \geq b > b'$ for all $i < k$. By definition of pointwise convergence topology, there exists $g \in A$ such that $g(x_{2i}) < a'$ and $g(x_{2i+1}) > b'$ for all $i < k$. Hence, $x \in D_k(A, E, a', b')$.]
(3) Any subset of a Talagrand μ-stable set is Talagrand μ-stable. (This is immediate from (1).)
(4) If A is Talagrand μ-stable, then so is $\overline{A}$. (This is immediate from (2).)

The main result of this section is that deep computations that are limit points of a Talagrand stable set of computations are Monte Carlo computable; this is Corollary 6.14 below. To prove it, we first show that the closure of a Talagrand μ-stable set is μ-measurable (lemma 6.11); but first, we need the following useful characterization of measurable functions (compare with Fact 1.2):

Fact 6.10 (Lemma 413G in [Fre03])**.** *Suppose that (X, Σ, μ) is a measure space and let $\mathcal{K} \subseteq \Sigma$ be a collection of measurable sets satisfying the following conditions:*

- *(X, Σ, μ) is complete, i.e., for all $E \in \Sigma$ with $\mu(E) = 0$ and $F \subseteq E$ we have $F \in \Sigma$.*
- *(X, Σ, μ) is semi-finite, i.e., for all $E \in \Sigma$ with $\mu(E) = \infty$ there exists $F \subseteq E$ such that $F \in \Sigma$ and $0 < \mu(F) < \infty$.*
- *(X, Σ, μ) is saturated, i.e., $E \in \Sigma$ if and only if $E \cap F \in \Sigma$ for all $F \in \Sigma$ with $\mu(F) < \infty$.*
- *(X, Σ, μ) is inner regular with respect to $\mathcal{K}$, i.e., for all $E \in \Sigma$*
$$\mu(E) = \sup\{\mu(K) : K \in \mathcal{K} \text{ and } K \subseteq E\}.$$

(In particular, if X is compact Hausdorff, $\mathcal{K}$ is the collection of compact subsets of X, μ is a Radon probability measure on X, Σ is the completion of the Borel σ-algebra by μ, and all these conditions hold). Then, $f : X \to \mathbb{R}$ is measurable if and only if for every $K \in \mathcal{K}$ with $0 < \mu(K) < \infty$ and $a < b$, either $\mu^*(K_{\leq a}) < \mu(K)$ or $\mu^*(K_{\geq b}) < \mu(K)$, where $K_{\leq a} = \{x \in K : f(x) \leq a\}$ and $K_{\geq b} = \{x \in K : f(x) \geq b\}$.

The following technical lemma will be instrumental for proving Proposition 6.13, which, in turn, will yield the main result of the subsection, namely Corollary 6.14

Lemma 6.11. *If A is Talagrand μ-stable, then $\overline{A}$ is also Talagrand μ-stable and $\overline{A} \subseteq \mathscr{M}^0(X, \mu)$.*

Proof. By Remarks 6.9, $\overline{A}$ is Talagrand μ-stable, and hence so is $\{f\}$ for every $f \in \overline{A}$. It therefore suffices to show that if $\{f\}$ is Talagrand μ-stable, then f is μ-measurable. If not, Fact 6.10 gives a set E of positive measure and $a < b$ with $\mu^*(E_{\leq a}) = \mu^*(E_{\geq b}) = \mu(E)$, where $E_{\leq a} = \{x \in E : f(x) \leq a\}$ and $E_{\geq b} = \{x \in E : f(x) \geq b\}$. Then $(E_{\leq a} \times E_{\geq b})^k \subseteq D_k(\{f\}, E, a, b)$ for every $k \geq 1$, and this product has outer measure $(\mu(E))^{2k}$ (see [Tal84]), so $\{f\}$ is not Talagrand μ-stable. $\square$

Definition 6.12. We say that A is *universally Talagrand stable* if A is Talagrand μ-stable for every Radon probability measure μ on X.

We first observe that universal Talagrand stability implies NIP:

Proposition 6.13. *Let X be a compact Hausdorff space and $A \subseteq \mathrm{C}(X)$ be pointwise bounded. If A is universally Talagrand stable, then A satisfies the NIP.*

Proof. By Lemma 6.11, $\overline{A} \subseteq \mathscr{M}^0(X, \mu)$ for every Radon probability measure μ on X, that is, $\overline{A} \subseteq M_r(X)$. By Theorem 6.5 (i)⇒(ii) with $K = X$, A satisfies the NIP. □

Corollary 6.14. *Let $(L, \mathcal{P}, \Gamma)$ be a CCS satisfying the Extendibility Axiom, let R be an exhaustive collection of sizers, and let $\Delta \subseteq \Gamma$ be R-confined. If $\pi_P \circ \Delta|_{L[r_\bullet]}$ is universally Talagrand stable for all $P \in \mathcal{P}$ and all $r_\bullet \in R$, then every deep computation in $\overline{\Delta}$ is universally Monte Carlo computable.*

Proof. This follows directly from Proposition 6.13 and Theorem 6.7. □

In the context of deep computations, we have identified two ways to obtain Monte Carlo computability, namely, NIP/PAC and Talagrand stability. It is natural to ask whether these two notions are equivalent. The following results show that, even in the simple case of countably many computations, this question is sensitive to the set-theoretic axioms. On the one hand, it is consistent (with respect to the standard ZFC axioms of set theory) that these two classes are the same:

Theorem 6.15 (Talagrand, Theorem 9-3-1(a) in [Tal84])**.** *Let X be a compact Hausdorff space and $A \subseteq M_r(X)$ be countable and pointwise bounded. Assume that $[0, 1]$ is not the union of $< \mathfrak{c}$ closed measure zero sets. If A satisfies the NIP, then A is universally Talagrand stable.*

(The assumption that $[0, 1]$ is not the union of $< \mathfrak{c}$ closed measure zero sets is a consequence of, for example, the Continuum Hypothesis.)

On the other hand, the opposite is also consistent for the particular case of the Lebesgue measure:

Theorem 6.16 (Fremlin, Shelah, [FS93])**.** *It is consistent with the usual axioms of set theory that there exists a countable pointwise bounded set of Lebesgue measurable functions with the NIP which is not Talagrand stable with respect to Lebesgue measure.*

Notice that the preceding two results apply to sets of measurable functions, a class of functions larger than the class of continuous functions. However, by the Extendibility Axiom, finitary computations are continuous, i.e., if A is a set of computations, then $A \subseteq \mathrm{C_p}(X)$. The example of Fremlin and Shelah therefore does not settle the question for computations, and we leave it open (Question 7.4).

7. Open questions

We collect the questions left open by this paper. The first concerns the hierarchy of Section 3.2, whose levels were defined through the topology of the space of deep computations.

Proposition 3.4 describes the three levels by the number of tests needed to tell deep computations apart, but these descriptions still involve infinitely many tests.

Question 7.1. Is there a finitary, combinatorial characterization of NIP_1, NIP_2 and NIP_3, in the spirit of the characterization of NIP by finite patterns in Theorem 3.1?

Theorem 6.7 shows that NIP suffices for Monte Carlo computability; whether it is also necessary is the next question.

Section 5 places a single iterated map at every level of the hierarchy except one: at the level NIP but not NIP_1, both examples are actions of groups with two generators (Section 5.5).

Question 7.2. Is there a cascade CCS that is NIP but not NIP_1? Equivalently, is there a continuous map γ of a compact metrizable space L such that (L, γ) is tame and $E(L, \gamma)$ is not first countable?

For actions of $\mathbb{Z}$, and more generally of abelian groups, the analogous question is Question 6.8 of Glasner and Megrelishvili [GM22].

Question 7.3. Under the assumptions of Theorem 6.7, suppose that every deep computation of Δ is universally Monte Carlo computable. Must $\pi_P \circ \Delta|_{L[r_\bullet]}$ have the NIP for all $P \in \mathcal{P}$ and all $r_\bullet \in R$?

Finally, Section 6 gives two routes to Monte Carlo computability, NIP and universal Talagrand stability, and the results of Talagrand and of Fremlin and Shelah leave open whether they can differ for computations.

Question 7.4. Is it consistent with the usual axioms of set theory that some countable, pointwise bounded set of continuous functions on a compact Hausdorff space satisfies the NIP but is not universally Talagrand stable? Equivalently, can the two routes to Monte Carlo computability of deep computations differ for countable sets of computations?

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